

Course Title: Computer Hardware Installation and Management

Course Code: UIL-CSC 202

Credit Unit: 2 Credits

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UIL-CSC 202: Computer Hardware Installation and Management

Introduction

Computer Hardware Installation and Management is a course designed to enhance students' knowledge in computer hardware installation, configuration, repair and their performance evaluation. The course will expose students to design of computer integrated circuits, internal component of computer system, coupling and decoupling a computer as well as troubleshooting and evaluating the performance of computer system. It will also equip student with the knowledge of operating system installation, configuration and management. This course provides the student with a prerequisite knowledge of computer hardware maintenance and repair and prepare them for a career in system maintenance and management.

Students will gain knowledge of the important concepts of computer hardware such as Integrated Circuit design and fabrication, ROM, RAM and hard disk installation, and their maintenance. The course will also include practical sessions where coupling, repair and configuration of hardware are demonstrated and how to evaluate the performance of a computer system and its optimization will also be demonstrated

Objectives

The objectives of the course are to:

- (i) describe the internal components of computer system to the students;
- (ii) describe the hardware memory devices of a computer system
- (iii) describe different kinds computer hardware circuits;
- (iv) explain and illustrate how to install major hardware components of computer system;
- (v) explain and experiment the installation of operating systems and other software on computer;
- (vi) discuss different integrated circuit design in computer system;
- (vii) explain and illustrate the installation of different peripheral devices on the computer system;
- (viii) explain and illustrate the troubleshooting processes of hardware components of the computer;
- (ix) discuss the performance evaluation of a computer system.
- (x) explain and demonstrate with practical's how to couple and uncouple a computer system.

Learning outcomes

At the end of this course students should be able to:

- (i) identify at least six (6) major internal components of computer system;
- (ii) identify at least three (3) memory hardware device of a computer system;
- (iii) identify at least three (3) kinds computer hardware circuits;

- (iv) install at least five (5) major hardware components of computer system;
- (v) install Windows operating systems computer;
- (vi) explain at least (2) two integrated circuit design in computer system;
- (vii) install at least four peripheral devices on the computer system;
- (viii) carry out simple troubleshooting process on hardware components of the computer;
- (ix) evaluate the performance of a computer system; and
- (x) perform the process of uncoupling and coupling of a desktop computer.

Course contents

Role of Computer unit in Organizations. Computer hardware circuits- diode arrays, PIAs etc. Integrated circuit fabrication process. Use of MIS, LIS, VLSI hardware design. Primary Memories- RAM, ROM. Secondary Memories- Hard disk, auxiliary, etc. Peripheral devices- Printers, CRT, Keyboards, Mouse, Scanners, etc. Integrated circuits and BUS. Operational amplifiers I- analog-to-digital converter. Operational amplifiers II- digital-to-analog converters. Computer hardware installation. Software installation. Configuration management. Computer Hardware maintenance. Computer Hardware repair. Computer security management. Computer system performance and evaluation. **Lab work:** Illustration on identification of hardware components of computer e.g. motherboard, , Processor, hard disk, RAM, ROM, VRAM, Integrated circuits and BUS. Coupling and decoupling of a computer system. Hands-on experiments on installation of internal components of computer system. Practicum on Operating system installation and formatting (Windows and Linux). Practicum on computer system configuration. Experiments on hardware fault troubleshooting and repair. Hands-on experiments on system performance evaluation

MODULE 1: INTRODUCTION TO COMPUTER HARDWARE

Unit 1: Role of Computer Systems in Organizations

1.0 Introduction

Computer systems play a vital role in modern organizations by enabling efficient data processing, communication, automation, and decision-making. Understanding their relevance is essential for effective hardware management.

Computers help in research, production, distribution, marketing, banking, team management, business automation, data storage, employee management, and very helpful to increase productivity at a lower cost, less time with high quality. That's why the **use** of the **computer** in any business organization cannot be overemphasized.

Computers have become essential business tools. They are **used** in every aspect of a company's operations, including product creation, marketing, accounting, administration and data processing.

It is critical that business owners take the time to choose the right **computers**, software and peripherals for their **organization**.

2.0 Learning Objectives

At the end of this unit, students should be able to:

- Explain the role of computer systems in organizations
- Identify areas where computers are applied

3.0 Main Content

3.1 Definition of Computer System

A computer system is made up of software programs and hardware components that have been carefully selected to perform well together.

The operating system (OS), which controls and offers services to other applications that can run on the computer, is the primary software component.

3.1.1 Role of Computers in an Organization

3.1.2 Introduction

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Computers have become essential business tools. They are used in every aspect of a company's operations, including product creation, marketing, accounting, administration and data processing. It is critical that business owners take the time to choose the right computers, software and peripherals for their organization.

3.1.3 Importance of Computers in an Organization

The advent of computers has revolutionized many industries, with office work being no exception. Computers have helped to improve the efficiency and accuracy of office workers by providing different software and communication functions to help execute work tasks. Computers are now a part of nearly every office, with most workplaces now unable to function without the use of computers.

The following are some of the importance of computer in an organization.

(i) Communication

Internal and external communication is much easier with the use of e-mail and internal messaging systems on computers. Office staff are able to pass information throughout the office quickly and

effectively, as most office setups have an alert system on individual computers when a message or e-mail is received. The Internet also greatly enhances communication options, with Skype and other such messaging and communication programs allowing national or transnational video and call conferencing with ease and less expense.

(ii) Data Storage

The data storage and retrieval capacity of computers is only getting greater and more advanced as technology improves. Files are easily retrievable through search functions, and hard drives can hold unprecedented volumes of files and data. For offices with large databases, such as governments, charities or other member-based associations, this data storage and retrieval function provides unparalleled advantages over traditional paper file storage, such as the ease and speed of information retrieval, the ease of changing data records and the ease of tracking changes made to customer, member or citizenry records.

3.1.4 The Role of Technology in Business Communication

Innovations in electronic communications technology spurred the Information Age, a time period featuring fluid, almost instantaneous information transmission. The digital world offers interfaces such as email, instant messaging and chat rooms that help with simultaneous transference of information to a large number of people. For many organizations, electronic communication has become a necessity for participation in modern commerce. Electronic commerce has opened up many opportunities because it allows large-scale global collaboration.

(i) Rapid Transmission

Electronic communication allows the rapid transfer of goods, money and ideas. This eliminates the need for employees to physically transfer ideas, through the mail, in person or over the phone. This also happens on a larger scale -- for example, sending a dozen email messages saves the worker the time it would have taken to make 12 phone calls or send 12 memos. To stay competitive, modern workplaces must work at the same speed as rivals, using the Internet and electronic communication.

(ii) Information Collaboration

Information is a valuable commodity to many organizations. Certain attributes, such as accuracy, accessibility, application and rarity, make information more valuable. Information that can be used to aid employees in decision-making is also valuable. The ability to communicate this information quickly adds to the value of the data. Employees who research and share are able to come to more informed decisions faster than if they used non-electronic communication.

3.1.5 The Advantages of Computers in the Office

Computer use is prevalent in many workplaces, and some companies may find functioning without them virtually impossible. As a result, computers have become an important part of how companies

conduct business as well as how workers perform their job tasks. Computers can even impact where employees do their jobs, as they can often work many miles from their official place of business.

(i) Improving Communication

Computers can improve communication both within the workplace and when dealing with customers. An e-mail can convey a large amount of information at one time and can be viewed at the convenience of the recipient. This can eliminate the sometimes-endless game of "phone tag" as well as the slowness of sending written information via postal mail. Computers also give customers an additional method of making inquiries or receiving information.

(ii) Improving Efficiency

Computers can increase the speed and accuracy of many work processes, which improves overall worker efficiency. Documents can be written and edited much more quickly with the aid of a word processing program, and procedures, such as billing and accounting, can also occur more rapidly and with fewer errors. Computers can produce reports with great speed and allow for the easy insertion of enhancements, such as charts, graphs and pictures if desired. Computers also help companies in administrative tasks, such as keeping up-to-date and accurate records.

3.1.6 The Impact of Social Media in the Workplace: Pros and Cons

Technology can help your business grow from a small, homegrown company to a large corporation. However, technology has a more immediate impact on your employees. Regardless of where you implement new technology in your organization, employees throughout your company will be directly affected in plenty of positive ways.

(i) Efficiency

Technology can improve the way your employees do their jobs, making them more efficient and free from the burden of tedious, repetitive tasks. Technology simplifies many job functions, which in turn strengthens performance and improves job satisfaction. For example, configuring Outlook to send personalized messages to a large group of email addresses can shave hours off the job of sending individual messages. Also, when used efficiently, technology improves accuracy – rather than typing and proofing individual messages, the sender can create one message, proof it and know that each addressee is receiving the same message. For more complex tasks, such as extensive reports, spreadsheets and presentations, technology can considerably reduce the time to create documents.

(ii) Communication

Remote employees benefit from communication technology. With VOIP (Voice Over Internet Protocol), employees who telecommute don't need to rely on their cell phones since the same type of multi-line office phone can be connected to an ethernet cord in a home office mirroring the

typical office setting, assuring remote employees they're part of a team. Future employees can attend job interviews that are outside reasonable commuting distance with video conferencing technology, which is particularly helpful for those who are looking to relocate.

3.1.2 Importance of hardware in system performance

Hardware is the foundational physical component of a computer system, dictating its maximum potential speed, efficiency, and reliability. High-performance hardware (CPU, RAM, SSD) enables faster data processing, improved multitasking, and faster storage access, preventing bottlenecks that slow down software applications.

Key aspects of hardware's importance in system performance include:

Processing Speed (CPU): The central processing unit's clock speed and core count determine how quickly the system executes instructions and runs applications.

Memory Capacity (RAM): Adequate RAM allows the system to hold more data for immediate access, enhancing multitasking and reducing slowdowns.

Storage Speed (SSD vs. HDD): Solid State Drives (SSDs) significantly improve system responsiveness, boot times, and data transfer speeds compared to traditional Hard Disk Drives (HDDs).

Graphics Processing (GPU): High-performance GPUs are essential for rendering high-quality visuals and speeding up compute-intensive applications.

Reliability and Upgradability: Modern, high-quality hardware ensures continuous operation, lowers the risk of failure, and allows the system to support future software requirements.

Efficient Bottleneck Management: Proper matching of components (e.g., matching a high-end CPU with fast RAM) ensures no single component limits the entire system's capability.

Without adequate hardware, even the best software cannot perform optimally, resulting in slow speeds and poor user experience.

4.0 Pilot Questions

- What is a computer system?
- List three roles of computers in organizations

5.0 Summary

In this unit, you have learnt that:

- Computer systems support organizational activities
- Hardware forms the backbone of system performance

6.0 Self-Assessment

- Explain the importance of computers in organizations
- List five applications of computer systems

7.0 Tutor Marked Assignment

- Discuss the role of computer systems in modern organizations

8.0 References

Unit 2: Internal Components of Computer System

1.0 Introduction

A computer system consists of several internal components that work together to process and store data.

2.0 Learning Objectives

Students should be able to:

- Identify major internal components
- Explain their functions

3.0 Main Content

3.1.0 What is Computer Hardware?

Hardware is the computer's peripheral devices which are used to assemble the system. Hardware devices majorly include computer input-output devices, memory, and CPU. Generally, Hardware is the internal and external components of a computer that allow you to accomplish major functions such as input, output, storage, communication, and processing.

Some most common examples of computer hardware are –

- (i) Input / Output devices like Keyboard, mouse, & monitor
- (ii) The central processing unit (CPU)
- (iii) Motherboard
- (iv) Random access memory (RAM)
- (v) Storage drive (HDD or SSD)
- (vi) Graphics processing unit (GPU)

Computer hardware is an umbrella term for computer peripherals. Hardware refers to tangible devices of the system that can physically see, touch and understand its structure. These are the essential components for a computer to work on. Computer hardware is attached to the system internally or externally, depending on whether the physical component is located within or outside of the system. Generally, internal hardware is essential for the computer to function, whereas external hardware improves its use or functionality. Internal computer hardware can be a video card, memory, or the central processing unit (CPU), whereas exterior computer hardware can include a keyboard or mouse.

3.1.1 Types of Computer Hardware

There are two types of computer hardware –

- (i) Internal hardware
 - (ii) External hardware
-
- (i) Internal Hardware

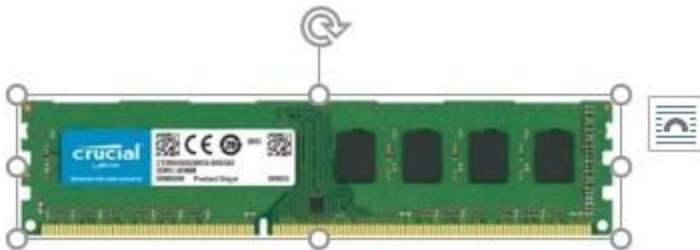
Internal hardware is the computer peripherals which are connected inside the computer. These include motherboards, hard drives, and DDR (RAM). The following images are the core examples of computer hardware.



Motherboard



Processor



RAM (Primary Memory)



Power Supply Unit

(ii) External Hardware

External hardware is the computer peripherals which are connected to the computer system externally. External hardware includes monitors, keyboards, printers, and scanners.



Keyboard



Mouse



Joystick



Monitor



Printer



3.1.2 Functions of Computer Hardware

Generally, functions of computer hardware depend on hardware units like input, output, and processing and storage units. Hardware functions in a computer system include a wide range of operations that allow the machine to complete its tasks efficiently. These functions are often separated into different categories, each handled by a unique set of hardware components.

Below are some of the key functions of the computer hardware's –

- **Input hardware** facilitates users to input the data into the system. A keyboard is used to type text, a Mouse to select an option by clicking on it, a scanner to scan images or pages etc.
- **Output hardware** produces the output into human understandable forms like a Monitor is used to display the output on the screen, a Printer to print output in the form of hard copy, speakers used to produce audio sound and projectors are used to project visual output on larger surfaces.
- **Processing hardware** like processors executes programs or instructions and performs calculations or does computations.
- **Storage hardware** like memory devices stores data to retain for future use. RAM for temporary storage and HDD to store data permanently.

- **Communication hardware** devices enable users to data exchange between multiple devices over the computer network. These include Network Interface Card (NIC), Modem, and Bluetooth and Wi-Fi Adapters.
- **Control hardware** manages and directs the operations of different components attached to the system. It includes a Motherboard, Power Supply Unit (PSU) and BIOS/UEFI.

3.2 The Motherboard

3.2.1 What is a Computer Motherboard?

A motherboard is a circuit board which is fixed inside the Central Processing Unit (CPU) of a computer. It can be considered as a computing system which integrates most of the essential components of a computer.

A motherboard serves as a single platform to connect all of the parts of a computer. It connects the CPU, memory, hard drives, optical drives, video cards, sound cards, and other ports and expansion cards directly or via cables. It can be considered as the backbone of a computer. Generally, it is made up of fibreglass and copper.



The motherboard is mounted inside the case and is securely attached via small screws through pre-drilled holes. The motherboard contains ports to connect all of the internal components. It provides a single socket for the CPU, whereas for memory, normally one or more slots are available. Motherboards provide ports to attach the floppy drive, hard drive, and optical drives via ribbon cables. The motherboard carries fans and a special port designed for power supply.

There is a peripheral card slot in front of the motherboard using which video cards, sound cards, and other expansion cards can be connected to the motherboard.

On the left side, motherboards carry several ports to connect the monitor, printer, mouse, keyboard, speaker, and network cables. Motherboards also provide USB ports, which allow compatible devices to be connected in a plug-in/plug-out fashion for example, pen drives, digital cameras, etc.

3.2.2 Components of a Motherboard

A motherboard is made up of different components; some of the primary elements are as follows –

- **CPU Socket** – Its a major component which determines the processor connected to the system.
- **Memory Slots** – These slots allow connecting memory devices into it.
- **Storage Connectors** – These slots include Serial ATA ports to connect hard drives like hard drives and SSDs.
- **ROM Slots** – These slots allow ROM BIOS to connect.
- **Chipset** – It connects CPU, memory, storage and peripheral devices.
- **Cooling fans** – Cooling fans that maintain a suitable internal operating temperature.
- **Peripheral connector** – these include USB ports to connect peripheral devices.
- **Audio Connectors** – These allow audio input and output devices to work with analogue and digital audio signals.
- **Network Connectors** – These ports are used for wired network connections, or Wi-Fi to access the internet.
- **Power connector** – power connectors are used to connect external power sources. There are two basic types: the 24-pin ATX power connector and the 4 or 8-pin CPU power connector.
- **BIOS/UEFI** – The Basic Input / Output System (BIOS), also known as the Unified Extensible Firmware Interface (UEFI), is a type of firmware that initializes and tests hardware during the boot process; it also provides runtime services to operating systems and programs.

3.3 Features of Computer Motherboard

A motherboard comes with the following features –

- Motherboard varies greatly in supporting various types of components.
- The motherboard supports a single type of CPU and a few types of memory.
- Video cards, hard disks, and sound cards have to be compatible with the motherboard to function properly.
- Motherboards, cases, and power supplies must be compatible to work properly together.

3.3.1 Types of Motherboards

Different types of Motherboards are as follows –

3.3.2 Advanced Technology (AT) motherboard

This motherboard was designed by IBM in 1984 with the IBM PC/AT. It was most widely used in industry and influenced the design of many subsequent motherboards. This motherboard has larger physical dimensions but is not suitable for smaller desktop computers.

The original AT motherboards were available in 12 inches 13.8 inches (305 mm x 351 mm) in size, making them relatively large by modern standards. The configuration of AT motherboards was not standardized, resulting in major differences between manufacturers. This frequently resulted in inadequate airflow and complex cable management.

The Advanced Technology (AT) motherboard was a key form factor in the history of personal computing, establishing standards that impacted subsequent designs. However, due to its large

size, convoluted layout, and restricted integration, it was eventually replaced by the more efficient and user-friendly ATX specification.

Standard ATX motherboard

ATX stands for advanced technology extended; it is an extended version of the AT motherboard that was created by Intel in the 1990s. It has become one of the most prevalent motherboard sizes used in desktop computers because of its compatibility and interchanged component features. A standard ATX motherboard is a popular desktop computer form factor that offers a good combination of size, capabilities, and expansion. Standard ATX motherboards are typically 305 mm x 244 mm (12 in x 9.6 in) in size. This size provides enough space for components and expansion slots.

ATX motherboard uses a common layout to enhance ventilation and component placement. The CPU socket is normally located toward the top or middle of the board, with expansion slots aligned for effective cooling and simplicity of installation.

3.3.3 Micro ATX motherboard

Micro ATX motherboards are smaller in size and have limited ports and slots than the Standard ATX board. This motherboard is better suited to users who don't desire too many connections and subsequent updates.

Micro ATX (mATX) motherboards are smaller versions of the conventional ATX motherboard form factor. It has been specifically designed concerning size and features which make it ideal for compact desktop builds while not compromising too much expandability or speed.

Micro ATX motherboards typically have dimensions of 244 mm x 244 mm (9.6 in x 9.6 in), which are smaller than conventional ATX but larger than Mini-ITX.

Micro ATX motherboards provide the right blend of size, capabilities, and expandability, making them a popular choice for a variety of desktop PC configurations. They provide ample PCIe slots and connectivity choices for most users while fitting in smaller cases than typical ATX boards. This makes them suited for a wide range of applications, including gaming, entertainment, and home office sets.

3.3.4 Extended ATX motherboard

An Extended ATX (eATX) motherboard is a larger variant of the normal ATX motherboard that is intended to provide additional features, improved performance, and expanded functionality like up to eight RAM slots, additionally, a higher number of PCIe (where e is for Express) and PCI slots.

An extended ATX (E-ATX) motherboard has a larger form factor than a conventional ATX motherboard, allowing for more components and more capabilities. E-ATX motherboards often have dimensions of 305 mm x 330 mm (12 in 13 in) or greater. They are broader than regular ATX boards, making sufficient space for more components and expansion slots.

E-ATX motherboards have a larger form factor than regular ATX, allowing for more components, expansion slots, and functionality. They are ideal for high-performance desktop builds and enthusiast systems that require flexibility, performance, and a wide range of networking options.

E-ATX motherboards typically have several PCIe slots, including several PCIe x16 slots for high-performance graphics cards and several PCIe x1 slots for further expansion cards.
Flex ATX motherboard

The FlexATX standard sets a balance between size and capability, making it a popular choice for projects that need a tiny footprint without sacrificing features. FlexATX motherboards are often smaller than normal ATX and microATX boards, with dimensions of 229 mm x 191 mm (9 in x 7.5 in).

Despite their compact size, FlexATX motherboards can provide a wide range of functionality such as multiple USB ports, integrated audio, and network connectivity. However, they may contain limited ports and connectors than larger form factors.

3.3.5 Low-Profile EXtended (LPX) motherboard

These are comparatively older motherboard form factors that were popular in the late 1980s and early 1990s. It was designed for low-profile desktop cases. The dimensions of LPX motherboards include around 330 mm x 229 mm (13 in x 9 in), although these can vary.

One of the distinguishing characteristics of LPX motherboards is the use of a riser card. The riser card enables expansion cards to be placed parallel to the motherboard, lowering the overall height of the system. This design was ideal for low-profile desktop cases. The LPX motherboard had a significant impact on the evolution of computer form factors, particularly in permitting the creation of smaller desktop systems. However, it has subsequently been replaced by more modern and adaptable standards.

3.3.6 BTX motherboard

BTX stands for balanced technology extended. It was designed by Intel in 2003 to overcome the limitations and thermal issues associated with the ATX standard. BTX motherboards are available in different sizes, including conventional, mini, and pico. Standard BTX boards typically measure 325 mm x 267 mm (12.8 in x 10.5 in). These have a different layout than ATX motherboards. They were created to optimize airflow and thermal management by arranging the key heat-generating components, such as the CPU and GPU, in a manner that allows for more efficient cooling. BTX motherboards include similar functionality and connectivity options to ATX boards, such as multiple USB ports, SATA connectors, integrated audio, and network interfaces.

3.3.6 Pico BTX motherboard

The Pico BTX (Balanced Technology eXtended) motherboard is a small version of the BTX form factor intended for ultra-compact devices. Pico BTX motherboards are available with 203 mm x 267 mm (8 in x 10.5 in) size. This compact size is designed for small form factor (SFF) cases.

Pico BTX boards, like all other BTX form factors, are designed to optimize airflow and thermal management. The CPU and other important heat-generating components are positioned to align with the system's airflow path, which typically runs from the front intake to the back exhaust.

3.3.7 Mini ITX motherboard

A Mini-ITX (Information Technology Extended) motherboard is a small form factor introduced by VIA Technologies in 2001. It was created for small form factor (SFF) computing, achieving a

balance of size, performance, and usefulness. Mini-ITX motherboards are 170 mm by 170 mm (6.7 in x 6.7 in) in size.

Despite its compact size, a Mini-ITX motherboard is designed to handle a wide range of components, making it appropriate for a wide range of computing requirements, including basic desktop use, gaming, and home theatre PCs.

3.3.8 Mini STX motherboard

In 2015, Intel developed the Mini-STX (Mini Socket Technology Extended) motherboard. It provides an ideal combination of size, performance, and upgradeability for small form factor (SFF) PCs. Mini-STX motherboards are 147 mm by 140 mm (5.8 in x 5.5 in) in size; the Mini-STX form factor is designed to fit in ultra-compact enclosures while yet delivering adequate performance for different applications.

Mini-STX motherboards offer a unique alternative for ultra-compact, upgradeable PCs. Their small size and socketed CPU make them excellent for SFF builds, home theatre PCs, and other applications where space is constrained but performance and upgradeability are still required.

Popular Motherboard Manufacturers

Following are the popular manufacturers of the motherboard.

- (i) Intel
- (ii) ASUS
- (iii) AOpen
- (iv) ABIT
- (v) Biostar
- (vi) Gigabyte
- (vii) MSI

4.0 Pilot Questions

- What is a motherboard?
- List four internal components

5.0 Summary

- Internal components perform processing and storage functions
- Each component has a specific role

6.0 Self-Assessment

- Explain the functions of CPU and RAM
- Differentiate between RAM and ROM

7.0 Tutor Marked Assignment

- Describe six internal components of a computer system

8.0 Reference

Unit 3: Graphics Processing Unit

1.0 Introduction

Graphics processing is an advanced technology that has evolved specifically for computing. The graphics processing unit (GPU) is an advanced and specialised processor which makes computing possibilities for machine learning, deep learning, gaming, content creation, and many more related technologies.

2.0 Learning Objectives

Students should be able to:

3.0 Main contents

3.1 What is Graphics Processing Unit (GPU?)

A graphics processing unit (GPU) is a specialised electrical circuit that accelerates computer graphics and image processing. GPUs are useful for non-graphic computations such as neural networks and cryptocurrency.

GPUs were originally developed to speed up the rendering of 3D visuals. They gradually became more adaptable and programmable and expanded their capabilities. This enabled graphics programmers to generate more fascinating visual effects and realistic scenes using advanced lighting and shadowing methods. In other applications, GPUs are most widely used to significantly work with high-performance computing (HPC), deep learning, and other domains.

3.2 Significances of GPUs

GPUs are mostly used to speed up real-time 3D graphics applications like gaming. However, as the twenty-first century began, computer experts recognised that GPUs could address some of the world's most complex computing problems.

This realisation ushered in the era of general-purpose GPUs. Graphics technology is now being used to solve a broader range of challenges. Today's GPUs are more programmable than ever before, allowing them to power a wide range of applications beyond standard graphics rendering.

GPUs are significant for the following reasons –

(a) GPUs for Gaming

Video games have become more computationally intensive, with hyperrealistic graphics and vast, complicated in-game worlds. With advanced display technologies, such as 4K screens and high

refresh rates, along with the rise of virtual reality gaming, demands on graphics processing are growing fast. GPUs are capable of rendering graphics in both 2D and 3D. With better graphics performance, games can be played at higher resolution, at faster frame rates, or both.

(b) GPUs for Video Editing and Content Creation

GPUs provide a computational solution for parallel processing, with built-in AI features and sophisticated acceleration for faster execution and easier rendering of video and graphics in higher-defined formats.

(c) GPU for Machine Learning

Artificial intelligence and machine learning are among the most interesting applications for GPU technology. Because GPUs have an enormous amount of computational power, they can provide remarkable acceleration in workloads that make use of GPUs' highly parallel nature, such as image recognition. Many of today's deep learning solutions rely on GPUs to work alongside CPUs.

3.3 How do GPUs work?

A GPU is ideally suited for tasks or programs that require large volumes of data and repetitive computation. GPUs have a similar fetch-decode-execute cycle, it has designed in such a way that they can handle thousands of threads at once, making them well-suited for parallel processing tasks



- **Fetching** – The GPU fetches instructions for the execution from its internal memory. (VRAM).
- **Decoding** – The control units of streaming multiprocessors (SMs) decode instructions. Each SM can process multiple instructions at once and assign them to available cores.
- **Executing** – The decoded instructions are executed by multiple small cores in the SM. These cores perform operations like matrix multiplication and vector operations for graphics rendering and machine learning.
- **Writing back** – The findings are either returned to GPU memory or transmitted to the display output.

Overall, GPU working includes –

- **Parallel Processing** – GPUs, unlike CPUs, have thousands of small cores which are integrated with the system to do parallel processing. This makes it extremely efficient for larger and more complex tasks that can be divided into smaller ones and apply parallel processing to them.
- **CUDA cores (NVIDIA) or stream processors (AMD)** – These are the basic processing units of a GPU, which can execute the processes on their own. Modern GPUs exist with thousands of these cores.
- **Memory** – A GPU has its memory which is known as Graphics Memory or VRAM (Video RAM). A VRAM is capable enough to do multiple tasks at once. This allows them to rapidly process and render images.

- **Shader** – Shaders are specialised processing units within a GPU. They are specifically designed to perform vertex shading, geometry shading, pixel shading, and computation. These units support rendering graphics and computations.
- **APIs** – To optimise GPUs, software applications use programming interfaces such as CUDA or OpenCL to manage and perform parallel activities. These APIs give support to developers with tools, frameworks and libraries to help them design programmes that make the best use of GPU capabilities.
- **Data and task parallelism** – GPUs are capable enough to perform data parallelism and task parallelism on different sets of data simultaneously.
- **SIMD (Single Instruction, Multiple Data)** – This paradigm enables GPUs to perform the same action on multiple data points at once making them useful for image processing and scientific computations.
- **Task distribution** – Workloads for graphics rendering or scientific simulations are divided into different smaller jobs; each of them is assigned to a separate thread that the GPU then runs concurrently. This strategy considerably reduces total processing time.
- **Thread execution** – Threads are smaller units of work that allow concurrent processing; this process shows how GPUs function. A GPU can support thousands of threads at a time.

3.4 Characteristics of GPU

Some of the key characteristics of GPU are as follows –

- **Parallel processing** – GPUs contain thousands of specialized to perform multiple tasks simultaneously. GPUs include SIMD, this feature allows the GPU to apply the same operation on multiple data points in parallel.
- **Memory** – GPUs have their memory. GPUs Video RAM enables rapid access to data and faster data transfer.
- **High Throughput** – The parallel processing feature of GPUs provides high throughput.
- **Shaders** – Shaders allow for transformations and lighting, texturing, shading and computational tasks beyond graphics rendering
- **Scalability** – GPUs can scale up that system from a few cores to thousands.
- **Software Ecosystem** – The GPU ecosystem supports APIs like as OpenGL, DirectX, and Vulkan, as well as frameworks like CUDA (NVIDIA) and OpenCL, allowing developers to use GPU capability for both graphics and general-purpose computing.

3.5 Applications of GPU

Some of the common application areas where GPUs are most widely used are as follows –

- **Gaming** – Gaming is most widely used for real-time rendering of high-resolution and complex graphics.
- **Visual effects** – It is used in films and television shows to produce realistic simulations, 3D animations, and intricate special effects,
- **3D Modelling** – CAD and 3D rendering applications fully rely on GPUs for 3D models.
- **AI and Machine Learning** – GPUs can process complex computations due to this; GPUs are used in AI and machine learning for training and inference of neural networks.
- **Scientific Computing** – GPUs are extremely useful in the domain of physics, chemistry, biology, simulations and data analysis.
- **Cryptocurrency Mining** – GPUs are used to validate transactions on blockchain networks, solving cryptographic puzzles.

- **Scientific Computing** – Scientific computing includes simulations and data analysis.

3.6 Advantages of GPUs

Some of the key advantages of GPUs are as follows –

- **Quicker execution** – GPU computing does fast execution and gives quicker results.
- **High processing power** – GPUs are made with thousands of small processing cores which perform tasks in parallel mode. This parallel processing enables a GPU to do multiple complex processing at a time.
- **Scalability** – GPUs are highly scalable.
- **Cost-effective solutions** – Graphics processing units are a cost-effective solution for a wide range of computing tasks, as they eliminate the need for specialised processors or hardware.
- **Graphics rendering** – High-quality 3D visuals and visual effects in animation, video games and simulations are rendered using GPU computing.

3.7 Disadvantages of GPUs

Some of the main disadvantages of GPUs are as follows –

- **Complexity** – Developing applications that properly utilise GPU capabilities is difficult and requires skills in specialised programming languages and frameworks.
- **Limited single-thread performance** – A small core of GPUs is less powerful than CPU cores for single-threaded tasks.
- **Memory constraints** – GPU memory is limited which can hamper tasks that require extensive memory access.
- **Compatibility and driver issues** – Updating GPU drivers is essential to optimise GPU performance.

4.0 Pilot Questions

5.0 Summary

6.0 Self-Assessment

7.0 Tutor Marked Assignment

8.0 Reference

MODULE 2

Unit 1: Power Supply Unit (PSU)

1.0 Introduction

2.0 Learning Objectives

Students should be able to:

3.0 Main Contents

3.1 Computer - Power Supply Unit (PSU)

1.1 What is a Power Supply Unit (PSU)?

A Power Supply Unit also known as PSU is an essential computer hardware component that converts alternating current (AC) into direct current (DC) and then supplies voltage to every component connected to the system. The power supply transforms a 110-115 or 220-230 volt AC to a stable low-voltage DC that the computer can use and is rated by the number of watts it generates.



Most of the power supplies are switched-mode (SMPS), which improves efficiency and simplifies design for different voltage inputs. PSUs have the abilities where the power input varies.

1.2 Why Power Supply Unit is Important?

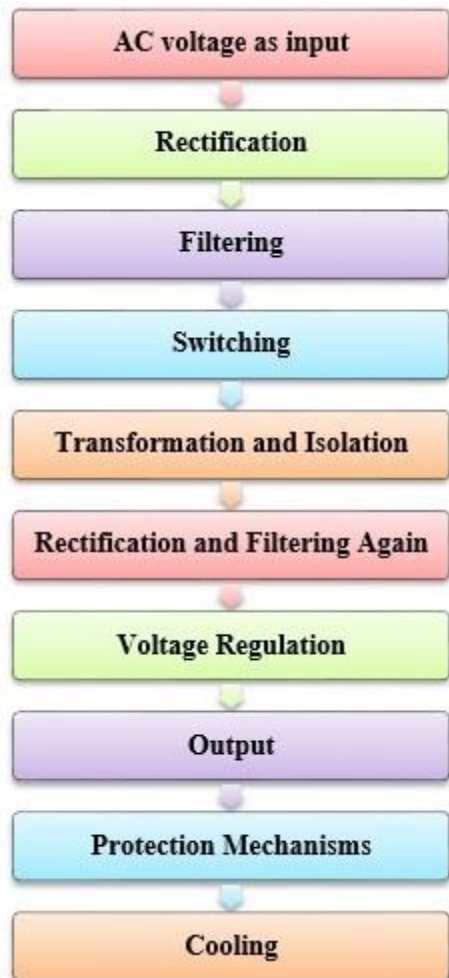
A power supply is needed and works as an essential component in the computer system for the following reasons –

- **Conversion of AC to DC Power** – A primary function of a PSU is to convert AC from the wall outlet into DC and make computers' important components functional like the motherboard, CPU, GPU, and storage drives because they operate on DC power.
- **Voltage Regulation** – A PSU supplies the required voltage to different components to ensure that they receive the required voltage to perform +12V, +5V, and +3.3V.

- **Power Distribution** – A PSU works as a power supply distributor for the different components connected to the system.
- **Protect system** – A PSU also protects systems from overvoltage short-circuit, and thermal protection.
- **Energy Efficiency** – Efficient power supplies can cut electricity costs and the environmental effects of running a computer.

1.3 How Does a Power Supply Unit (PSU) Work?

A Power Supply Unit works by converting AC to DC and supplying required voltages to the different components connected to the system. Following are the steps to demonstrate how a PSU works –



- **AC voltage as input** – The PSU takes AC voltage as input.
- **Rectification** – A rectifier converts inputted AC voltage into DC.
- **Filtering** – The rectified DC power is subsequently filtered by capacitors to smooth out fluctuations and remove ripple, producing a more stable DC voltage.
- **Switching** – Switching is a process of sending filtered DC power to a high-frequency transformer; the switch-mode power supply (SMPS) converts DC voltage to high-frequency AC. This high-frequency AC enables the use of a smaller and more efficient transformer.

- **Transformation and Isolation** – The high-frequency AC is then transferred using a transformer, which steps down the voltage to the desired levels. This transformer keeps isolation between input and output to improve safety.
- **Rectification and Filtering Again** – The transformed AC voltage is rectified back into DC with diodes. This DC is then filtered once again with capacitors to remove any residual ripple, resulting in get stable DC voltage.
- **Voltage Regulation** – Voltage regulators are used to keep output voltages within limits, despite the changes in input voltage. This is critical to the computer's stable operation.
- **Output** – The regulated DC voltages are supplied to every component connected to the system using specified connectors and cables. It includes common voltages like +12V, +5V, and +3.3V.
- **Protection Mechanisms** – Modern PSUs protect the system from overvoltage, short circuits, and thermal protection.
- **Cooling** – PSUs are equipped with cooling fans which keep systems cool from system heat generated during operations.

1.4 Types of Power Supply Units

Types of Power Supply are as follows –

- **ATX Standard** – These are the most common PSUs usually used in desktop systems. These power supplies are tailored to be compatible with the ATX motherboard and deliver three distinct positive voltage rails: +3.3 V, +5 V, and +12 V, in addition to a standby voltage rail of +5 V SB, which ensures that computers receive power when in standby mode.
- **Entry-Level Power Supply Specification (EPS)** – Based on the ATX standard, this PSU was specifically designed for servers. It sets a more stable environment for critical applications than ATX standards, which makes it perfect for critical servers.
- **Small Form Factor (SFX)** – These PSUs are used in smaller form-factor computers using MicroATX motherboards. These are the power supply units used in smaller set-top boxes like DVD players and cable boxes.
- **Thin Form Factor (TFX)** – TFX power supplies are longer and thinner than traditional ATX power supplies which are used in ultra-slim and low-profile desktops.
- **Modular and Semi-Modular Power Supply** – Modular PSUs allow users to connect using cables; this process minimises cable clutter and improves airflow while Semi-Modular PSUs use fixed and detachable cables.
- **Non-Modular Power Supply** – All cables are permanent and not removable. These are frequently less expensive, however, they result in more cable clutter.
- **Laptop Power Supply / AC Adapter** – External power supplies that convert AC to DC to power laptop computers. These are referred to as power bricks or AC adapters.
- **Uninterruptible Power Supply (UPS)** – An external device that provides backup power during power outages, rather than a computer's power supply unit. It contains batteries and can keep a system working for a short period if the main power fails.

1.5 Functions of a Power Supply Unit

The Power Supply Unit (PSU) in a computer performs different essential functions. Some of them are as follows –

- **Power Conversion** – The main function of a PSU is to convert AC from the wall outlet into DC. Generally, it provides multiple DC voltage outputs like +12V, +5V, +3.3V.

- **Voltage Regulation** – A PSU supplies the required voltage to different components to ensure that they receive the required voltage to perform +12V, +5V, and +3.3V. This is crucial for reliable operation and stable voltage.
- **Power Distribution** – A PSU works as a power supply distributor to the different components connected to the system like motherboard, CPU, graphics card, storage devices, and peripherals.
- **Protection Mechanisms** – It protects the system from overvoltage short-circuit, thermal protection, and over-temperature.
- **Efficiency** – PSUs reduce power wastage and reduce electricity costs.
- **Cooling** – It supports system cooling by heat dissipation and temperature management.
- **Standby Power** – The PSU can provide standby power (often +5VSB) to support features such as Wake-on-LAN and allow the system to execute specific tasks during low power mode or when the system is turned off.

1.6 How to Choose a Power Supply Unit (PSU)?

When choosing the right Power Supply Unit (PSU) for your computer; the following considerations are required –

- **Determine required power watt** – You should estimate the total wattage needed for the system like CPU, GPU, motherboard, RAM, storage drives, and any peripherals. A user should also keep in mind about future upgrades for the system.
- **Efficiency** – Efficient PSUs can lower electricity costs.
- **Connectors and Cables** – It determines that the PSU has the required connectors and pins.

Unit 2: Computer – Software

1.0 Introduction

2.0 Learning Objectives

At the end of this unit, students should be able to:

3.0 Main Contents

3.1 Computer Software

Software is a set of programs, which is designed to perform a well-defined function. A program is a sequence of instructions written to solve a particular problem.

There are two types of software –

- System Software
- Application Software

3.1.2 System Software

The system software is a collection of programs designed to operate, control, and extend the processing capabilities of the computer itself. System software is generally prepared by the computer manufacturers. These software products comprise of programs written in low-level languages, which interact with the hardware at a very basic level. System software serves as the interface between the hardware and the end users.

Some examples of system software are Operating System, Compilers, Interpreter, Assemblers, etc.



Here is a list of some of the most prominent features of a system software –

- Close to the system
- Fast in speed
- Difficult to design
- Difficult to understand
- Less interactive
- Smaller in size
- Difficult to manipulate
- Generally written in low-level language

3.1.3 Application Software

Application software products are designed to satisfy a particular need of a particular environment. All software applications prepared in the computer lab can come under the category of Application software.

Application software may consist of a single program, such as Microsoft's notepad for writing and editing a simple text. It may also consist of a collection of programs, often called a software package, which work together to accomplish a task, such as a spreadsheet package.

Examples of Application software are the following –

- Payroll Software
- Student Record Software
- Inventory Management Software
- Income Tax Software
- Railways Reservation Software
- Microsoft Office Suite Software
- Microsoft Word
- Microsoft Excel
- Microsoft PowerPoint



3.1.3 Features of application software are as follows –

- Close to the user
- Easy to design
- More interactive

- Slow in speed
- Generally written in high-level language
- Easy to understand
- Easy to manipulate and use
- Bigger in size and requires large storage space

3.2 Software Types

Software is a collection of programs, data, and instructions that enable a computer system and its hardware to do the tasks. A program is a set of instructions or programming statements.

Hence, software built using multiple statements. It is an important part of modern computing and is classified into different categories based on its capability and purpose. For example, word-processing software enables the user to create, edit and save documents. A web browser enables the user to view and share web pages and multimedia files.

Software Types

Some common types of software are as –

- System Software
- Application Software
- Utility Software
- Open Source Software

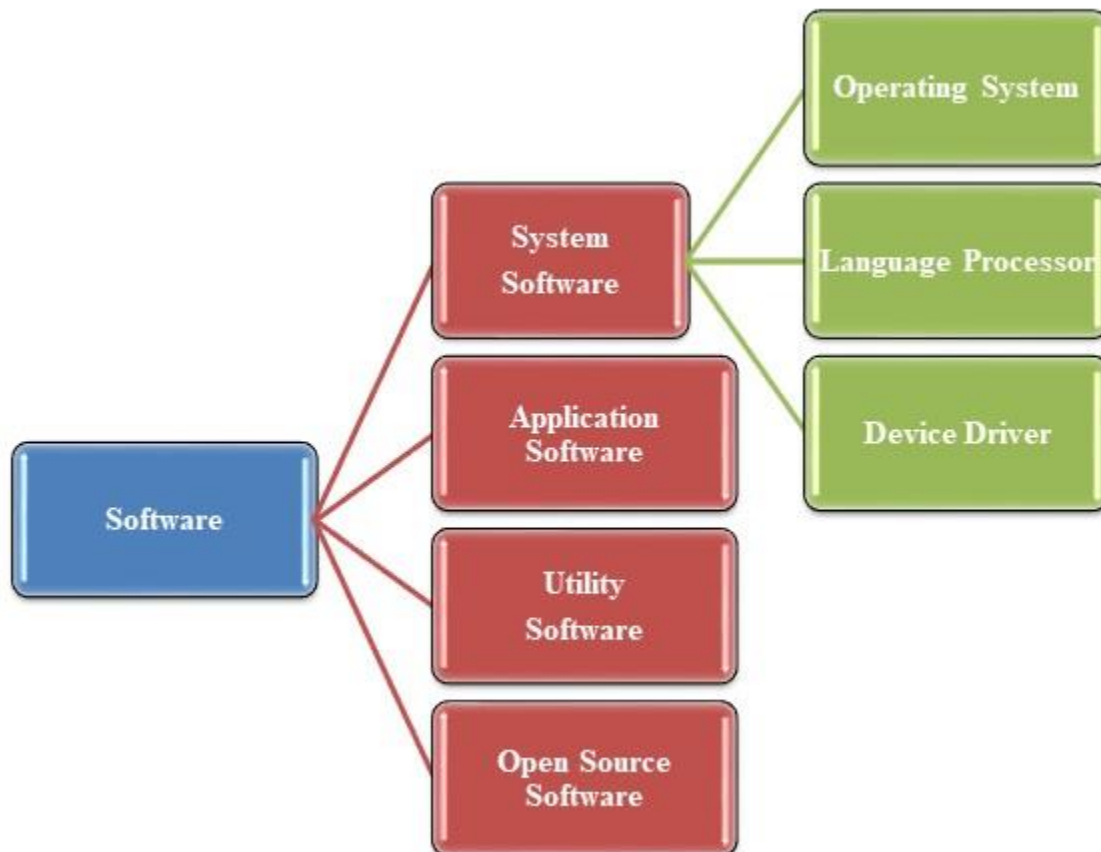


Figure 1.4 Software and its types

(i) System Software

Software that enables a computer to run and activates required hardware or other parts of a computer to run is called **system software**. System software acts as **interface** between hardware and user applications. An interface is needed because hardware devices or machines and humans speak in different languages.

Machines understand only binary language i.e. 0 (absence of electric signal) and 1 (presence of electric signal) while humans speak in English, French, German, Tamil, Hindi and many other languages. English is the pre-dominant language of interacting with computers.

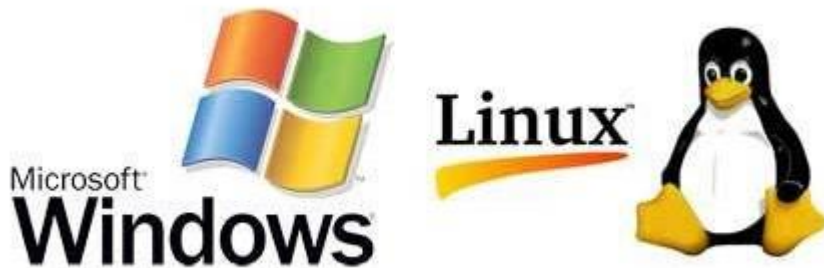
Software is required to convert all human instructions into machine understandable instructions. And this is exactly what system software does.

Based on its function, system software is of four types –

- Operating System
- Language Processor
- Device Drivers

Operating System

Generally, system software is responsible for functioning of all hardware parts and their interoperability to carry out tasks successfully is called operating system (OS). OS is the first software to be loaded into computer memory when the computer is switched on and this is called booting.



OS manages a computer's basic functions like storing data in memory, retrieving files from storage devices, scheduling tasks based on priority, etc.

Language Processor

As discussed earlier, an important function of system software is to convert all user instructions into machine understandable language. When we talk of human machine interactions, languages are of three types –

- **Machine-level language** – This language is nothing but a string of 0s and 1s that the machines can understand. It is completely machine dependent.
- **Assembly-level language** – This language introduces a layer of abstraction by defining mnemonics. Mnemonics are English like words or symbols used to denote a long string of 0s and 1s. For example, the word "READ" can be defined to mean that computer has to retrieve data from the memory. The complete instruction will also tell the memory address. Assembly level language is machine dependent.

- **High level language** – This language uses English like statements and is completely independent of machines. Programs written using high level languages are easy to create, read and understand.

Program written in high level programming languages like Java, C++, etc. is called source code. Set of instructions in machine readable form is called object code or machine code. System software that converts source code to object code is called language processor. There are three types of language interpreters –

- **Assembler** – Converts assembly level program into machine level program.
- **Interpreter** – Converts high level programs into machine level program line by line.
- **Compiler** – Converts high level programs into machine level programs at one go rather than line by line.

Device Drivers

System software that controls and monitors functioning of a specific device on computer is called device driver. Each device like printer, scanner, microphone, speaker, etc. that needs to be attached externally to the system has a specific driver associated with it. When you attach a new device, you need to install its driver so that the OS knows how it needs to be managed.

(ii) **Application Software**

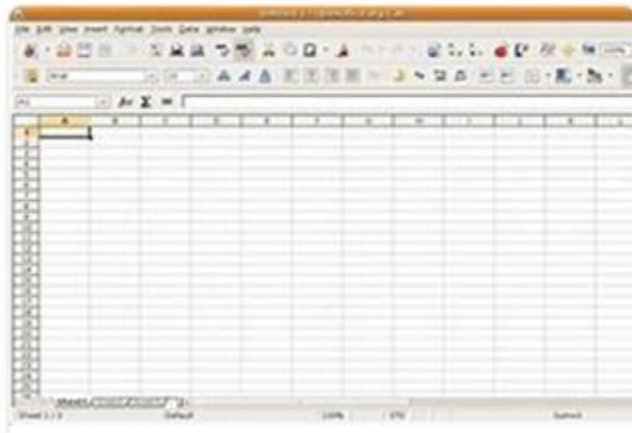
Application software, sometimes known as "apps" or "applications," is a type of software that is created for certain tasks or functions such as productivity, entertainment, communication, or other reasons. Hence, software that performs a single task and nothing else is called **application software**. Application software is very specialized in their function and approach to solving a problem. So spread sheet software can only do operations with numbers and nothing else. Application software, as opposed to system software (such as operating systems and device drivers), is designed to fulfil the needs and preferences of end users. The following are some examples of common types of application software –

Productivity Software

- **Word Processing** – This application software allows users to create, edit, and format documents. Some popular word processing applications are Microsoft Word, Google Docs, and Apple Pages.



- **Spreadsheet Application** – This application software is used to create and analyzing spreadsheets. Some popular Spreadsheet applications are Microsoft Excel, Google Sheets, and LibreOffice Calc.



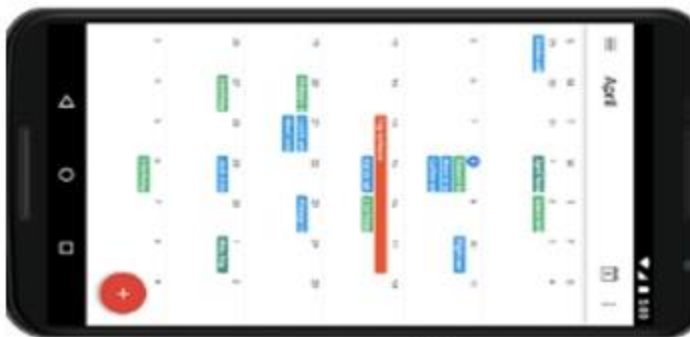
- **Presentation Software** – This application software is used to make slideshows and presentations. Some popular presentation applications are Microsoft PowerPoint, Google Slides, and Apple Keynote.



- **Project Management Software** – This application software is used to make teams plan, organize, and track projects. Examples - Trello, Asana, and Jira.



- **Calendar and Scheduling Apps** – This application software is used for time management and scheduling appointments, like Google Calendar, Microsoft Outlook, and Apple Calendar.



Graphics and Multimedia Software

- **Image Editing** – This application is used to edit and manipulate images. Examples are Adobe Photoshop, GIMP, and Canva.
- **Video Editing** – This application is used to create and editing of video. Examples are Adobe Premiere Pro, Final Cut Pro, and DaVinci Resolve.
- **Audio Editing** – This application is used for recording and editing audio. Examples are Audacity and Adobe Audition.
- **3D Modelling and Animation** – This application is used for creating 3D graphics and animations. Examples are Blender and Autodesk Maya.

Communication Software

- **Email Clients** – This application is used to send, receive, and manage email. Examples are Microsoft Outlook, Mozilla Thunderbird, and Apple Mail.
- **Instant Messaging Apps** – This application enables users for real-time text and multimedia communication. Examples are WhatsApp, Slack, and Telegram.
- **Video Conferencing Software** – This application is used to conduct virtual meetings and conferences. Examples are Zoom, Microsoft Teams, and Cisco Webex.
- **Social Media Apps** – These applications provide platforms for social networking. Examples are Facebook, Twitter, and Instagram.

Web Browsers

These application programs are used to access and navigate websites on the internet. Popular web browsers are Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

Gaming Software

These applications are designed for entertainment and gaming. Examples are Steam, PlayStation, Xbox, and mobile app stores.

(iii) Utility Software

Provides various utilities to perform specific tasks, such as –

- **File Compression and Extraction** – Examples are WinZip, 7-Zip, and WinRAR.
- **Backup and Recovery** – Examples are Acronis True Image and Time Machine.
- **Disk Cleanup and Optimization** – Examples are CCleaner and Disk Cleanup (Windows).

Education and E-Learning Software

These applications are designed to learning management systems (LMS); automate educational system, online education, and digital libraries.

Health and Fitness Apps

These applications are designed to keep track of health metrics, Yogas and Asans, dietary guidance etc.

Financial Software

These applications are specifically designed to manage personal finances, accounting, and taxation. Examples are QuickBooks, TurboTax, and Mint.

Travel and Navigation Apps

These applications are designed to assist travellers with trip planning, maps, and GPS navigation etc. Examples are Google Maps, TripAdvisor, and Waze.

In continuation of changing demands and preferences of users in different fields, application software continues to improve and diversify. Users can install and utilise these applications to do certain tasks and enhance their digital experiences on their computers, smartphones, tablets, and other digital devices.

(iv) **Utility Software**

Application software also known as utility programs or system utilities that assists system software in doing their work is called utility software. Utility software is intended to help users to manage and maintain their computer systems, as well as optimise performance and execute different system-related activities. These programmes are often used to assure the efficient operation of hardware and software components, and improve a computer system's overall functionality and stability. Thus utility software is actually a cross between system software and application software.

Some common types of utility are as –

- **Antivirus and Antimalware Software** – These utilities keep safe the computer against malicious software, spyware, and viruses, as well as other types of security threats. Examples are Norton Antivirus, McAfee, and Malwarebytes.
- **Disk Cleanup and Optimization** – These utilities help to remove temporary or unwanted files. For example - CCleaner (Crap Cleaner) removes temporary files, cache data, and unused files to free up disk space. Defragmentation applications reorganise fragmented files on a hard drive in order to increase read/write access.
- **Disk Partitioning and Management** – Disc Management (Windows) and GParted (Linux) are utilities that enable users to create, resize, and manage partitions on their hard drives.
- **Backup and Recovery** – Backup utilities automatically create copies of vital data to prevent loss. Example: Windows Backup and Time Machine (Mac). Data recovery software recovers erased files and folders. Popular options include Recuva and TestDisk.
- **System Monitoring and Diagnostic Tools** – Task Manager (Windows) and Activity Monitor (Mac) offer real-time system resource monitoring to find and fix performance issues. Windows Memory Diagnostic and Apple Diagnostics (Mac) diagnose hardware issues.
- **Driver Updaters** – For hardware compatibility and performance, these programmes update out-dated or incompatible device drivers. Examples are Driver Booster and Driver Easy.
- **Password Managers** – Users generate, store, and manage complex, secure passwords with password management software. Examples are LastPass, Dashlane, and 1Password.
- **Firewall Utilities** – A firewall protects a computer or network by managing incoming and outgoing network traffic. Examples are Windows Firewall and third-party firewall solutions like ZoneAlarm.
- **Data Encryption Software** – These utilities encrypt sensitive data to protect it from unauthorized access. Examples are BitLocker (Windows) and FileVault (Mac).
- **Uninstaller Programs** – These utilities assist users in thoroughly removing undesirable apps as well as the files and registry entries linked with those applications. Examples are Revo Uninstaller and IObit Uninstaller.

(v) **Open Source Software**

Software whose **source code** is freely distributed with a license to study, change and further distributed to anyone for any purpose is called **open source software**. Open source software is generally a team effort where dedicated programmers improve upon the source code and share the changes within the community. Open source software provides these advantages to the users due to its thriving communities –

- Security
- Affordability
- Transparent
- Interoperable on multiple platforms
- Flexible due to customizations
- Localization is possible

Freeware

A software that is available free of cost for use and distribution but cannot be modified as its source code is not available is called freeware. Examples of freeware are Google Chrome, Adobe Acrobat PDF Reader, Skype, etc.

Shareware

Software that is initially free and can be distributed to others as well, but needs to be paid for after a stipulated period of time is called shareware. Its source code is also not available and hence cannot be modified.

Proprietary Software

Software that can be used only by obtaining license from its developer after paying for it is called proprietary software. An individual or a company can own such proprietary software.

Its source code is often closely guarded secret and it can have major restrictions like –

- No further distribution
- Number of users that can use it
- Type of computer it can be installed on, example multitasking or single user, etc.

For example, Microsoft Windows is a proprietary operating software that comes in many editions for different types of clients like single-user, multi-user, professional, etc.

4.0 Pilot Questions

5.0 Summary

In this unit, you have learnt that:

6.0 Self-Assessment

7.0 Tutor Marked Assignment

8.0 References

MODULE 3 - Data Storage & Memory

Unit 1 Computer - Data Storage & Memory

3.0 Main Contents

3.1 Data Storage & Memory

Data storage and memory are both important parts of computers, but they do different things when it comes to storing and retrieving information. Computer memory stores data and instructions for the CPU to process tasks. Memory is vital for computer efficiency and comes in many forms.

1.1 Data Storage – The permanent, long-term storing of digital information is referred to as data storage. Data is stored on different physical and virtual storage systems and media. There are several types of data storage –

Primary Storage (Main Memory or RAM)

Computers keep active data in primary storage. Data is lost when the machine is turned off since it is volatile memory.

Data and programmes in use are stored quickly and temporarily. RAM is volatile, losing data when the computer is turned off. A computer with greater RAM can multitask and run apps smoothly.

Secondary Storage (Non-volatile Storage)

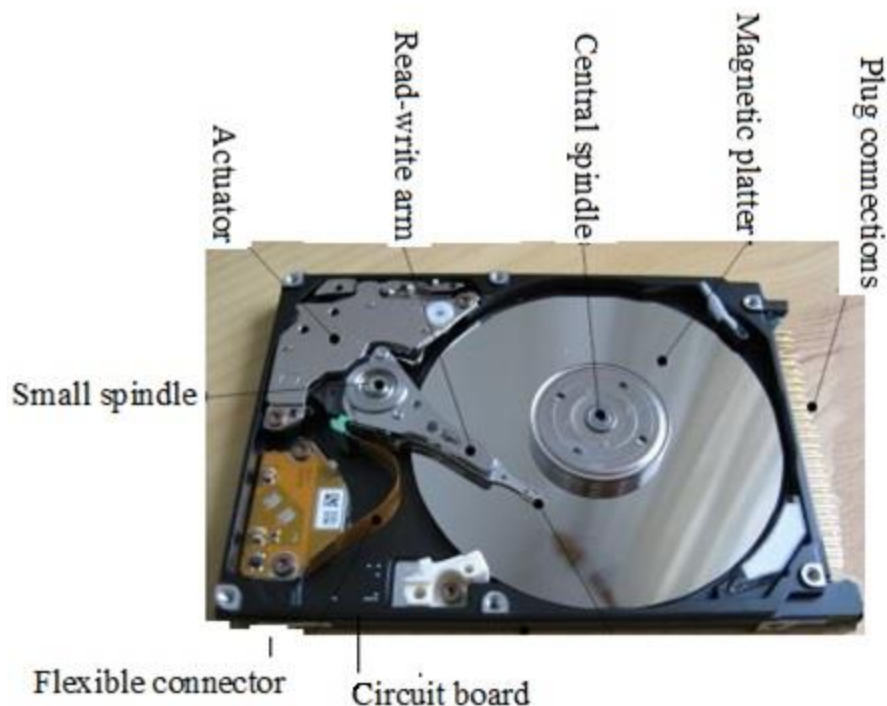
Secondary storage holds long-term data. It stores data even when the machine is off, unlike primary storage. HDDs, SSDs, CDs, DVDs, and USB drives are examples.

1.2 Hard Disk Drives (HDD)

The hard disc drive is a main hardware in a computer that stores data. Most files, including the operating system and programme titles, are kept on the hard disc drive.

1.3 HDD and its components

A hard drive has few fundamental elements. One or more shiny silver platters store magnetically stored information, an arm mechanism moves a tiny magnet called a read-write head over the platters to record or store information, and an electronic circuit controls everything and connects the hard drive to your computer.



- **Actuator** – The actuator arm is a mechanical element responsible for providing support to the read/write heads. The mechanism facilitates the movement of the read/write heads to the intended track on the platter for the purpose of data reading or writing. Contemporary hard disc drives (HDDs) employ voice coil actuators to achieve precise positioning.
- **Read-write Heads** – Every platter is equipped with a corresponding read/write head that hovers slightly above its surface. These components are tasked with the responsibility of retrieving data from and storing data onto the platters. The actuator arm is utilised to swiftly traverse the surface of the platter, facilitating access to various sections of the disc.
- **Central Spindle** – Platters rotate at a consistent pace with the spindle. Most HDDs spin at 5,400 or 7,200 RPM, while some high-performance devices spin at 10,000 or 15,000 RPM.
- **Magnetic Platter** – it stores information in binary form. The flat, round discs inside an HDD are called platters. Most of the time, these plates are made of aluminium or glass and have a magnetic coating. Magnetic patterns keep the information on the platters.
- **Power Connectors** – It links hard drive to circuit board in personal computer. A power connector is needed to power internal HDDs from the computer's power source. Modern discs use SATA power connectors.
- **Read-Write Head** – is a tiny magnet on the end of the read-write arm.
- **Circuit Board** – The controller board, also called a "Printed Circuit Board" (PCB), is the HDD's brain. It holds the drive's software, which controls how data is accessed and fixes errors. The drive is also connected to the computer's interface (such as SATA or IDE) by the driver board.
- **Data Connector** – The computer's power supply device powers internal HDDs through a power connector. SATA power connectors are common for newer drives.
- **Small Spindle** – This allows read-write arm to swing across platter.

Platters are the most critical hard drive elements. Their name implies that they are hard discs made of glass, ceramic, or aluminium coated with a thin metal coating that may be magnetised or demagnetized. Although compact hard drives have one platter, each side is magnetically coated. Larger drives feature platters stacked on a spindle with a tiny gap. The read-write heads can reach any region of the platters, which rotate at up to 10,000 rpm.

A five-platter hard drive would need ten read-write heads because each platter contains two, one for the top and one for the bottom. An electrically operated arm moves the read-write heads from the drive centre to the edge and back.

1.4 Features of HDDs

- **Data Storage** – HDDs consist of one or more rigid, rapidly rotating disks (platters) coated with a magnetic material. Data is stored on these platters in the form of magnetic patterns. To read or write data, a read/write head moves over the spinning platters.
- **Capacity** – HDDs are available in various storage capacities, ranging from a few gigabytes (GB) to multiple terabytes (TB). The capacity of an HDD is determined by the number of platters it contains and their data density.
- **Speed** – HDDs are generally slower than SSDs in terms of data access times and read/write speeds. This is because the read/write heads need to physically move to the correct location on the disk, causing latency.
- **Durability** – HDDs are mechanical devices with moving parts, which makes them susceptible to mechanical failure if subjected to physical shocks, drops, or excessive vibration. SSDs, being solid-state, are more durable in this regard.
- **Price** – HDDs are typically less expensive per gigabyte compared to SSDs. This makes them a cost-effective choice for large storage needs.
- **Noise and Heat** – HDDs can generate noise due to the spinning platters and moving heads. They also generate heat, so proper ventilation is necessary to prevent overheating.
- **Power Consumption** – HDDs generally consume more power than SSDs because of the mechanical components. This can lead to slightly higher electricity usage in laptops and desktops.
- **Longevity** – The lifespan of an HDD can vary but is often shorter than that of an SSD. Over time, the mechanical parts can wear out, leading to data loss.
- **Applications** – HDDs are commonly used for various purposes, such as storing large files (photos, videos, and documents), running operating systems and software, and in network-attached storage (NAS) devices for backup and data sharing.
- **Backup** – Due to their mechanical nature, it's important to regularly back up data on HDDs to prevent loss in case of failure. Redundant storage systems or RAID configurations can be used for added data protection.

1.5 How hard drives work ?

The working of an HDD depends on its components which are described above. The main elements; a spinning platter and an actuator arm which are as:

- **Magnetic Platters** – Platters refer to circular plates. The number of platters in a hard drive is directly proportional to its storage capacity, as each platter can accommodate a specific quantity of information. Consequently, a hard drive with greater storage capacity would include a larger number of platters compared to one with lesser storage capacity. The

process of storing and retrieving information from the platters involves the utilisation of concentric circles known as tracks, which are further divided into sectors.

- **Arm** – The arm refers to the component protruding above the platters. The arms equipped with read and write heads that are utilised for the purpose of reading and storing magnetic data onto the platters. Each platter will possess an own arm that is utilized for the purpose of reading and writing data from and onto it.

The motor is employed to rotate the discs at speeds ranging from 4,500 to 15,000 revolutions per minute (RPM). A higher rotational speed of a drive corresponds to improved performance outcomes. When a computer requires data retrieval from the hard drive, the motor initiates the rotation of the platters, while the arm repositions itself to the designated location above the platter where the data is stored. The magnetic heads located on the arm are responsible for detecting the magnetic bits present on the platters. These heads then convert the detected bits into the corresponding data, which can be utilized by the computer. In contrast, during the process of data transmission to the drive, the magnetic heads will emit magnetic pulses towards the platters, so altering the magnetic characteristics of the platter surface and subsequently storing the information.

HDDs are cheaper and have a higher capacity than SSDs, but they are slower and less durable. The use case and cost-performance-storage capacity balance determine whether to utilise HDDs or SSDs. Computers and electronics store and retrieve data on HDDs. Based on their speed and dependability, Solid State Drives (SSDs) have become popular after decades of use.

4.0 Pilot Questions

5.0 Summary

In this unit, you have learnt that:

6.0 Self-Assessment

7.0 Tutor Marked Assignment

8.0 References

Unit 2: Solid State Drives

1.0 Solid State Drives (SSD)

A Solid State Drive (SSD) is a data storage device commonly employed in computers and various electronic gadgets. In contrast to conventional Hard Disc Drives (HDDs), which employ rotating discs for data retrieval and storage, Solid State Drives (SSDs) utilise NAND-based flash memory technology. Solid-state drives (SSDs) have experienced a surge in popularity owing to their different advantages which encompass enhanced speed, reduced energy usage, and heightened resilience.



Solid-state drives (SSDs) are capable of permanently storing data within an integrated circuit, commonly utilising flash memory technology. The utilisation of flash memory within a SSD facilitates the electrical and noiseless processes of data writing, transferring, and erasing. Unlike mechanical hard-disk drives (HDDs), SSDs do not have any moving components. Solid-state drives offer great performance and low noise levels due to their lack of moving components. However, it is important to note that SSDs are generally more expensive than HDD.

In the past, SSDs possessed a more restricted storage capacity in comparison to conventional hard disc drives. However, presently, both SSDs and HDDs are available in a wide range of sizes to cater to diverse storage requirements. SSDs are frequently employed in premium computing systems or as supplementary storage components within personal computers intended for consumer use.

1.1 How an SSD works

- **NAND Flash Memory** – The primary component of an SSD is NAND flash memory. This is a type of non-volatile storage that retains data even when the power is turned off. NAND flash memory is made up of memory cells organized into pages and blocks.
- **Cells and Bits** – Each memory cell in NAND flash can store multiple bits of data, typically one, two, or three bits per cell. The more bits a cell can store, the more cost-effective the SSD is, but it can also affect performance and longevity.
- **Pages and Blocks** – Data is written and read from NAND flash memory in fixed-size pages, which are grouped into larger blocks. When data is written to an SSD, it is first programmed into an empty page. To update or delete data, SSDs use a process called wear-leveling, which ensures that data is evenly distributed across the NAND cells to prevent excessive wear on any one cell.
- **Controller** – The SSD controller is a critical component that manages the reading and writing of data to and from the NAND flash memory. It handles tasks such as error correction, wear leveling, and garbage collection to optimize performance and longevity.
- **TRIM** – The TRIM command is an important feature of SSDs. It allows the operating system to inform the SSD which data blocks are no longer in use, allowing the SSD to perform efficient garbage collection and wear leveling, which helps maintain performance over time.
- **Wear Leveling** – NAND flash memory cells have a limited number of write-erase cycles before they degrade. Wear leveling ensures that data is written evenly across the memory cells, extending the lifespan of the SSD.

- **Read and Write Operations** – SSDs can read data very quickly since there are no moving parts involved. Write operations, however, can be more complex because data must be written to empty pages and previously used pages must be erased before they can be rewritten. This process is managed by the SSD controller to optimize speed and minimize write amplification.

SSDs work by storing data in NAND flash memory cells, using a controller to manage read and write operations, and implementing various techniques like wear leveling and garbage collection to ensure longevity and maintain performance. Their speed, reliability, and efficiency have made them a preferred choice for storage in modern computers and electronic devices.

1.2 Tertiary Storage

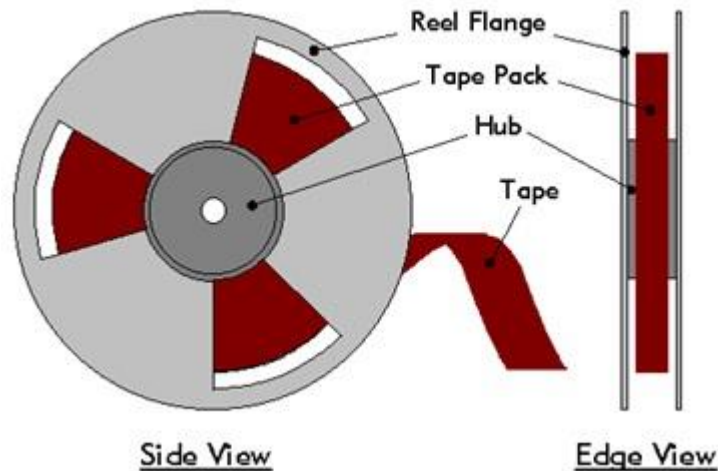
Tertiary storage medium is used for long-term backups and archiving. It's slower and less used than main and secondary storage but offers high-capacity storage for long-term data.

Magnetic tape, optical discs, and specialised hard drives are tertiary storage media.

1.3 How tertiary storage work?

Magnetic Tape

- **Recording Mechanism** – Magnetic tape is a long, thin plastic strip covered with magnets. Magnetising regions representing binary 0s and 1s stores data on tape.
- **Reading Mechanism** – As tape runs through a tape drive, magnetised portions are read to recover data. This sequential procedure requires the drive to fast forward or rewind to access specific data, which is slower than random access storage like hard drives or SSDs.



Magnetic tape has a small layer that records magnetic signals and a thicker film backing. The top coat, or magnetic coating, is a polymer-bound magnetic pigment. Magnetic particles and tape backing are held together by the binder. Magnetic impulses are stored in the top coat, or magnetic layer.

1.4 Optical Discs

Optical discs read and write data using optical technology. They are flat, circular polycarbonate discs with a reflective data layer and a protective outer layer.

Different types of digital data are stored on optical discs –

- **CD (Compact Disc)** – CDs were one of the earliest types of optical discs and were released in the early 1980s. They can store up to 700 MB of data and are widely used for music, software, and data storage.
- **DVD (Digital Versatile Disc)** – After CDs, DVDs have substantially higher storage capacity, 4.7 GB single-layer and 8.5 GB dual-layers. In movies, software distribution, and data backup.
- **Blu-ray Disc** – Newer optical discs like Blu-rays can store more data than DVDs. They can hold 25 GB (single-layer) and 50 GB (dual-layer) for standard discs and more for specialised versions. Blu-ray discs are utilised for HD video and data storage.

- **Archival Disc** – Archival optical discs preserve data for long periods. Archival-grade data volume and durability make them ideal.
- **Recordable and Rewritable Discs** – Rewritable and write-once optical discs are available. Rewritable discs can be erased and rewritten, however recordable discs can only be written once.

1.5 How Optical discs work?

Optical discs work by using a laser to read and write data.

- **Recording Mechanism** – Lasers create tiny pits and lands on CDs, DVDs, and Blu-rays. Pits and lands are binary data.
- **Reading Mechanism** – The disc drive's laser detects pits and lands' reflections to read data. Drive converts reflections into digital data.



A less powerful laser is used to read the digital information from the designs on the reflective layer. To write data, a more powerful laser heats up a small part of the reflective layer, making pits and bumps that indicate binary data.

A popular way to store data and distribute video, optical discs now have to compete with other storage technologies like USB drives, external hard drives, and cloud storage. But they are still used in some situations where their durability, ability to store data even when the computer is not on, and resistance to data corruption are important.

1.6 USB Flash Drives

A USB flash drive, also known as a thumb drive, jump drive, pen drive, or USB memory stick, is a storage device that can be used to record information on a very small flash memory chip. Data can be read from it and saved by users. These storage devices are meant to be considerably more compact than a normal storage disc; in fact, some of them are about the size of a thumb. USBs can connect to any computer; because of the compatibility of their universal serial bus, often known as a USB, port.

Data can be stored on USB drives, which are portable storage devices. Because of their portability and the ease with which they can transfer and store data, they are very compact and have a low

weight. As a result of these characteristics, they have become an extremely common storage device.



Security Flash Drive



Music Flash Drive



Credit Card Hard Drive



Keychain USB Flash Drive

1.7 Features of USB flash drives

- **Storage Capacity** – USB flash drives are available in a wide range of storage capacities, spanning from a few gigabytes (GB) to several terabytes (TB).
- **Physical Size** – These devices are generally characterised by their small and compact size, resembling a plastic or metal stick with a USB connector located at one end. These devices are specifically engineered for convenient portability, allowing them to be effortlessly carried in a pocket or securely attached to a keychain.
- **Data Transfer Speed** – USB flash drives exhibit varying data transfer speeds, typically classified according to USB generations, including USB 2.0, USB 3.0, USB 3.1, and USB 3.2. Every successive generation presents enhanced data transfer rates. USB 3.0 and subsequent iterations exhibit notable enhancements in speed compared to USB 2.0.
- **File Storage** – A USB flash drive is capable of storing a wide range of data types, including documents, photos, videos, music, software, and other files. These devices are frequently utilised for the purpose of transferring files between various devices or serving as a reliable backup storage solution.
- **Plug-and-Play** – USB flash drives are commonly recognised as plug-and-play devices, which implies that they can be effortlessly connected to a USB port on a computer and are immediately operational, without necessitating the installation of supplementary drivers or software.

- **Security** – Certain USB flash drives are equipped with integrated security features, including hardware encryption and password protection, which serve to enhance the protection of your data in the event of loss or theft of the drive.
- **Price** – The pricing of USB flash drives is subject to variation depending on factors such as storage capacity, speed, and brand. In general, drives with smaller capacities are typically more cost-effective, whereas drives with larger capacities and faster speeds tend to be priced at a higher range.

Unit 3: Memory Cards

1.0 Introduction

Memory cards are a type of storage device that can hold data files. It gives you a place to store data and files from the connected device. Hence, a memory card is a form of storage media utilised for the purpose of storing various types of data, such as photos, videos, or other electronic information, within electronic devices. Commonly utilised devices that make use of a memory card encompass digital cameras, digital camcorders, handheld computers, MP3 players, PDAs, cell phones, game consoles, and printers.

Some common types of memory cards are as –

- **Secure Digital (SD) Card** – SD cards are popular memory cards. MicroSD, miniSD, and normal SD are available. Digital cameras, smartphones, and other portable devices require SD cards.



- **CompactFlash (CF) Card** – SD cards are smaller and weaker than CompactFlash cards. Professional cameras and industrial applications use them.



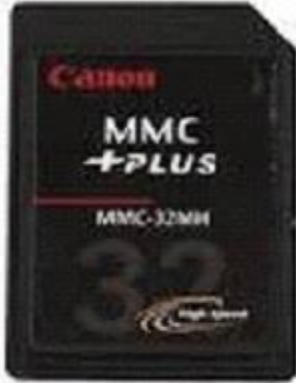
- **Memory Stick** – Memory Stick cards were popular in Sony cameras, camcorders, and laptops.



- **XQD and CFexpress** – These high-speed memory card formats are utilised in professional cameras and video recorders. They can capture high-resolution photos and videos due to their quick read and write rates.



- **MultiMediaCard (MMC)** – MMC cards are smaller than SD cards and used in older devices. Today they're uncommon.



- **SmartMedia Card** – Most digital cameras and other portable devices no longer use SmartMedia cards.



Data transfer speeds and storage capacity vary by memory card. Memory cards should be compatible, have enough storage, and be fast enough for their intended usage. Choose a memory card that fits your needs, such as high-speed video recording or data transfer.

Since memory cards are vulnerable to harsh handling and extreme circumstances, they should be handled carefully to avoid physical damage and data corruption. Data loss can be avoided by regularly backing up memory card data to external storage media.

1.1 Specialized Hard Drive Systems

Specialised hard drive systems are storage solutions that are made for specific applications. They are often made to fit the needs of a specific industry or job. These systems are different from consumer or business hard drives you can buy off the shelf.

They are usually customised to improve speed, reliability, and features.

Here are some examples of specialized hard drive systems –

- **Network Attached Storage (NAS)** – NAS hard drives are designed for household and small business file sharing and storage. Multiple storage bays, RAID functionality, and built-in network connectivity make file access and sharing easy.
- **Storage Area Network (SAN)** – Multiple servers receive block-level storage from high-performance SANs using specialised hard drive systems. Data centres and enterprises utilise them to centralise and manage storage resources.
- **Video Surveillance Storage** – Video surveillance systems need hard drives for continuous HD video recording and playback. These drives are optimised for write-intensive workloads and surveillance.
- **Gaming Storage** – Some consoles and high-end PCs include specialised hard drives for fast loading and smooth gaming. SSDs minimise load times on these discs.
- **Ruggedized Hard Drives** – Durable hard drives can survive extreme temperatures, stress, and vibration. Industries like military, aerospace, and industrial automation employ them.
- **Data Recovery Systems** – To recover data from damaged or corrupted drives, data recovery businesses utilise specialised hard drive systems with modern tools and software.
- **Medical Imaging Storage** – Medical imaging technologies like MRI and CT scans create massive amounts of data that hospitals need to store. Specialised storage solutions protect data and comply with privacy laws.
- **Digital Content Creation (DCC) Storage** – DCC professionals like video editors and 3D animators need high-capacity, high-speed storage for huge multimedia files. These needs are met by specialised storage systems.
- **Scientific Research Storage** – Scientific research, notably genomics and particle physics, requires huge data storage and analysis. Effective big data jobs require specialised storage solutions.

Application-specific hard drive systems often provide features and optimisations not present in consumer-grade hard drives. To fulfil industry or use case needs, these solutions may feature redundancy, fault tolerance, data integrity measures, and specialised interfaces.

1.2 Features of Tertiary Storage

- **Media Types** – Magnetic tapes, optical discs, and archival media are used in tertiary storage systems. These inexpensive media can hold lots of data.
- **Low Access Speed** – Tertiary storage accesses data slower than primary and secondary storage. To read tape data, you must physically move it.
- **High Capacity** – Terabytes or petabytes are typical storage capabilities for tertiary storage systems. Long-term data retention is possible with them.
- **Archival and Backup** – Archival data is held in tertiary storage for lengthy periods and accessed rarely. Additionally, it is utilised for disaster recovery data backup.
- **Reliability** – Tertiary storage media can preserve data for long periods without degradation due to its durability and longevity.

- **Offline Storage** – Offline or near-line tertiary storage keeps media from rotating or being instantly accessible. They are loaded or mounted as needed.
- **Cost-Effective** – Tertiary storage is cost-effective for massive, infrequently accessed data. Media are cheaper than high-speed, high-capacity disc drives.

Tertiary storage is beneficial in healthcare (for patient records), banking (for past transaction data), and government (for archive papers) since data retention regulations are strict. It provides an efficient and cost-effective way to store and protect large amounts of data. However, its slower access times make it unsuitable for real-time transaction processing and database searching. These applications are unsuitable.

1.3 Cloud Storage

Cloud storage is a method of storing data on remote servers that are administered and operated by a service provider that is not directly involved with the user. Typically, the retrieval and administration of this data occur via internet connectivity, as opposed to being stored on local storage devices such as hard drives or on-site servers. Cloud storage services have numerous benefits, such as scalability, accessibility, cost-effectiveness, and data redundancy.

Features of cloud storage

- **Scalability** – Users are able to adjust the amount of data they store in the cloud to meet their requirements because to the cloud's high degree of scalability. This elasticity is advantageous for firms that have storage requirements that are subject to change.
- **Accessibility** – If you have access to the internet, you can retrieve data saved on the cloud from anywhere. Because of this, it is much simpler for individuals and organisations to access their data using a variety of devices, such as personal computers, smartphones, and tablets.
- **Cost-effectiveness** – Cloud storage generally charges users for the storage capacity and resources they use. This may be cheaper than updating on-premises storage.
- **Security** – The security of user data is a top priority for cloud storage providers. This consists of access controls, frequent security audits, and the encrypting and decrypting of data at all times. Users must take responsibility for the security and configuration of their data in the cloud.
- **Backup and Disaster Recovery** – Cloud storage services have backup and restore tools built in. Users can set up regular backups and recover data in case they lose data or the system breaks down.



When choosing a cloud storage provider, organizations and individuals should consider factors such as their specific storage needs, budget, security requirements, and the provider's reliability

and reputation. Different cloud storage services may offer various features and pricing structures, so it's important to select one that aligns with your unique requirements.

Data storage holds data for long-term storage, while RAM holds data and instructions for immediate CPU execution. Computers need storage for capacity and memory for speed and response.

4.0 Pilot Questions

5.0 Summary

In this unit, you have learnt that:

6.0 Self-Assessment

7.0 Tutor Marked Assignment

8.0 References

MODULE 4: Computer - Types of Data Storage

Unit 1: Types of Data Storage

1.0 What is Data Storage?

The permanent, long-term storing of digital information is called data storage. Data is stored on different physical and virtual storage systems and media.

Data storage uses magnetic, optical, or mechanical mediums to record and maintain data/information to store, retrieve and process it. Data storage and memory are important parts of computers, but they do different things when storing and retrieving information. Computer memory stores data and instructions for the CPU to process tasks. Memory is vital for computer efficiency and comes in many forms.

1.1 How does Data Storage Work?

Generally, in modern computers, the storage devices are connected to the system directly or using a network. Users send requests to the system to access and save data on different storage devices. A method of collecting data and stored into the system plays a vital role in the data storage, accessing and processing.

1.2 Forms of Data Storage

Mainly, there are three ways to store the data –

- **File Storage** – This form of data storage is a hierarchical form that organizes and stores data.
- **Block Storage** – Block storage is another data storage form that organizes and stores data into blocks.
- **Object Storage** – Object storage is also known as object-based storage which is used to organise and store large amounts of unstructured data.

1.3 Types of Data Storage

Majorly data storage is categorised into two parts; described below –

1. Primary Storage (Main Memory or RAM)

Computers keep active data in primary storage. Data is lost when the machine is turned off since it is a volatile memory. Data and programmes in use are stored quickly and temporarily. RAM is volatile, losing data when the computer is turned off. A computer with greater RAM can multitask and run apps smoothly.

2. Secondary Storage (Non-volatile Storage)

Secondary storage holds long-term data. It stores data even when the machine is off, unlike primary storage. HDDs, SSDs, CDs, DVDs, and USB drives are examples.

3. Tertiary Storage

Tertiary storage medium is used for long-term backups and archiving. It's slower and less used than main and secondary storage but offers high-capacity storage for long-term data. Magnetic tape, optical discs, and specialised hard drives are tertiary storage media.

1.4 Features of Tertiary Storage

Some of the primary features of Tertiary Storage are as follows –

Feature	Description
Media Types	Magnetic tapes, optical discs, and archival media are used in tertiary storage systems. These inexpensive media can hold lots of data.
Low Access Speed	Tertiary storage accesses data slower than primary and secondary storage. To read tape data, you must physically move it.
High Capacity	Terabytes or petabytes are typical storage capabilities for tertiary storage systems. Long-term data retention is possible with them.
Archival and Backup	Archival data is held in tertiary storage for lengthy periods and accessed rarely. Additionally, it is utilised for disaster recovery data backup.
Reliability	Tertiary storage media can preserve data for long periods without degradation due to its durability and longevity.
Offline Storage	Offline or near-line tertiary storage keeps media from rotating or being instantly accessible. They are loaded or mounted as needed.
Cost-Effective	Tertiary storage is cost-effective for massive, infrequently accessed data. Media are cheaper than high-speed, high-capacity disc drives.

Tertiary storage is useful in healthcare, banking, and government, where data retention laws are stringent. It offers an efficient and cost-effective solution for storing and protecting large amounts of data. However, its slower access times render it unsuitable for real-time transaction processing and database searching.

4.0 Pilot Questions

5.0 Summary

In this unit, you have learnt that:

6.0 Self-Assessment

7.0 Tutor Marked Assignment

8.0 References

Unit 2: Computer - Hard Disk Drives (HDD)

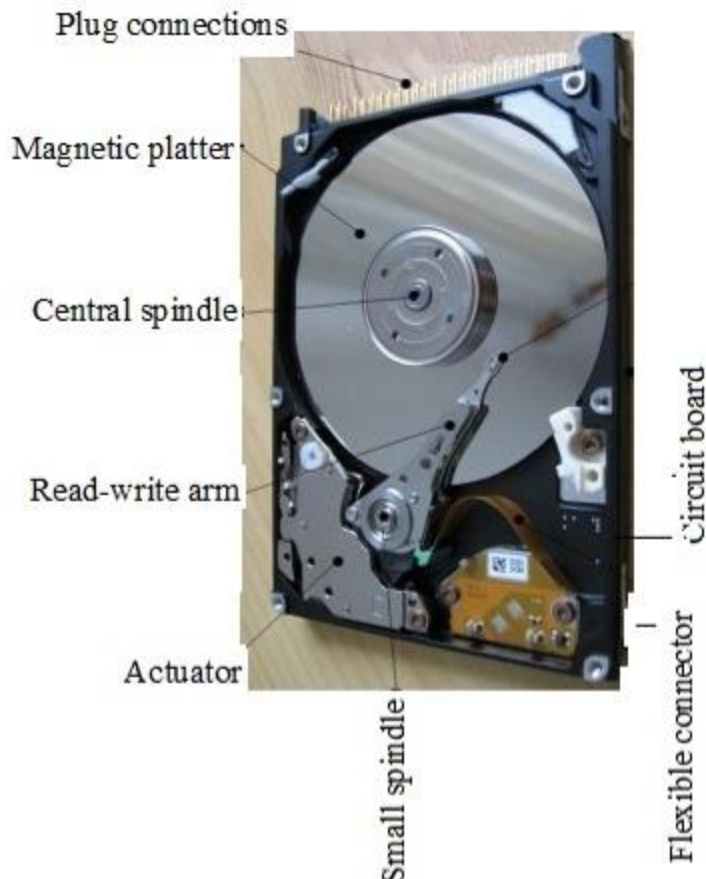
3.0 Main Contents

3.1 What are Hard Disk Drives (HDD)?

The hard disc drive is the main hardware in a computer that stores data. Most files, including the operating system and programme titles, are kept on the hard disc drive.

1.1 Components of Hard Disk Drives

A hard drive has a few fundamental elements. One or more shiny silver platters store magnetically stored information, an arm mechanism moves a tiny magnet called a read-write head over the platters to record or store information, and an electronic circuit controls everything and connects the hard drive to your computer.



1. Actuator

The actuator arm is a mechanical element responsible for providing support to the read/write heads. The mechanism facilitates the movement of the read/write heads to the intended track on the platter for data reading or writing. Contemporary hard disc drives (HDDs) employ voice coil actuators to achieve precise positioning.

2. Read-write Heads

Every platter is equipped with a corresponding read/write head that hovers slightly above its surface. These components are tasked with the responsibility of retrieving data from and storing data on the platters. The actuator arm is utilised to swiftly traverse the surface of the platter, facilitating access to various sections of the disc.

3. Central Spindle

Platters rotate at a consistent pace with the spindle. Most HDDs spin at 5,400 or 7,200 RPM, while some high-performance devices spin at 10,000 or 15,000 RPM.

4. Magnetic Platter

it stores information in binary form. The flat, round discs inside an HDD are called platters. Most of the time, these plates are made of aluminium or glass and have a magnetic coating. Magnetic patterns keep the information on the platters.

5. Power Connectors

These link the hard drive to the circuit board in a personal computer. A power connector is needed to power internal HDDs from the computer's power source. Modern discs use SATA power connectors.

6. Read-write Head

The read-write head is a tiny magnet on the end of the read-write arm.

7. Circuit Board

The controller board, also called a "Printed Circuit Board" (PCB), is the HDD's brain. It holds the drive's software, which controls how data is accessed and fixes errors. The drive is also connected to the computer's interface (such as SATA or IDE) by the driver board.

8. Data Connector

The computer's power supply device powers internal HDDs through a power connector. SATA power connectors are common for newer drives.

9. Small Spindle

This allows the read-write arm to swing across the platter. Platters are the most critical hard drive elements. Their name implies that they are hard discs made of glass, ceramic, or aluminium coated with a thin metal coating that may be magnetised or demagnetized. Although compact hard drives have one platter, each side is magnetically coated. Larger drives feature platters stacked on a spindle with a tiny gap. The read-write heads can reach any region of the platters, which rotate at up to 10,000 rpm.

A five-platter hard drive would need ten read-write heads because each platter contains two, one for the top and one for the bottom. An electrically operated arm moves the read-write heads from the drive centre to the edge and back.

1.2 Features of Hard Disk Drives

Some of the key features of Hard Disk Drives or HDDs are as follows –

Data Storage

HDDs consist of one or more rigid, rapidly rotating disks (platters) coated with a magnetic material. Data is stored on these platters in the form of magnetic patterns. To read or write data, a read/write head moves over the spinning platters.

Capacity

HDDs are available in various storage capacities, ranging from a few gigabytes (GB) to multiple terabytes (TB). The capacity of an HDD is determined by the number of platters it contains and their data density.

Speed

HDDs are generally slower than SSDs in terms of data access times and read/write speeds. This is because the read/write heads need to physically move to the correct location on the disk, causing latency.

Durability

HDDs are mechanical devices with moving parts, which makes them susceptible to mechanical failure if subjected to physical shocks, drops, or excessive vibration. SSDs, being solid-state, are more durable in this regard.

Price

HDDs are typically less expensive per gigabyte compared to SSDs. This makes them a cost-effective choice for large storage needs.

Noise and Heat

HDDs can generate noise due to the spinning platters and moving heads. They also generate heat, so proper ventilation is necessary to prevent overheating.

Power Consumption

HDDs generally consume more power than SSDs because of the mechanical components. This can lead to slightly higher electricity usage in laptops and desktops.

Longevity

The lifespan of an HDD can vary but is often shorter than that of an SSD. Over time, the mechanical parts can wear out, leading to data loss.

Applications

HDDs are commonly used for various purposes, such as storing large files (photos, videos, and documents), running operating systems and software, and in network-attached storage (NAS) devices for backup and data sharing.

Backup

Due to their mechanical nature, it's important to regularly back up data on HDDs to prevent loss in case of failure. Redundant storage systems or RAID configurations can be used for added data protection.

1.3 How do Hard Disk Drives (HDDs) Work?

The working of an HDD depends on its components which are described above. The main elements; a spinning platter and an actuator arm are as –

Magnetic Platters

Platters refer to circular plates. The number of platters in a hard drive is directly proportional to its storage capacity, as each platter can accommodate a specific quantity of information. Consequently, a hard drive with greater storage capacity would include a larger number of platters compared to one with lesser storage capacity. The process of storing and retrieving information from the platters involves the utilisation of concentric circles known as tracks, which are further divided into sectors.

Arm

The arm refers to the component protruding above the platters. The arms are equipped with read and write heads that are utilised to read and store magnetic data onto the platters. Each platter will possess its arm that is utilised to read and write data from and onto it.

The motor is employed to rotate the discs at speeds ranging from 4,500 to 15,000 revolutions per minute (RPM). A higher rotational speed of a drive corresponds to improved performance outcomes. When a computer requires data retrieval from the hard drive, the motor initiates the rotation of the platters, while the arm repositions itself to the designated location above the platter where the data is stored. The magnetic heads located on the arm are responsible for detecting the magnetic bits present on the platters. These heads then convert the detected bits into the corresponding data, which can be utilised by the computer. In contrast, during the process of data transmission to the drive, the magnetic heads will emit magnetic pulses towards the platters, so

altering the magnetic characteristics of the platter surface and subsequently storing the information.

HDDs are cheaper and have a higher capacity than SSDs, but they are slower and less durable. The use case and cost-performance-storage capacity balance determine whether to utilise HDDs or SSDs. Computers and electronics store and retrieve data on HDDs. Based on their speed and dependability, Solid State Drives (SSDs) have become popular after decades of use.

4.0 Pilot Questions

5.0 Summary

In this unit, you have learnt that:

6.0 Self-Assessment

7.0 Tutor Marked Assignment

8.0 References

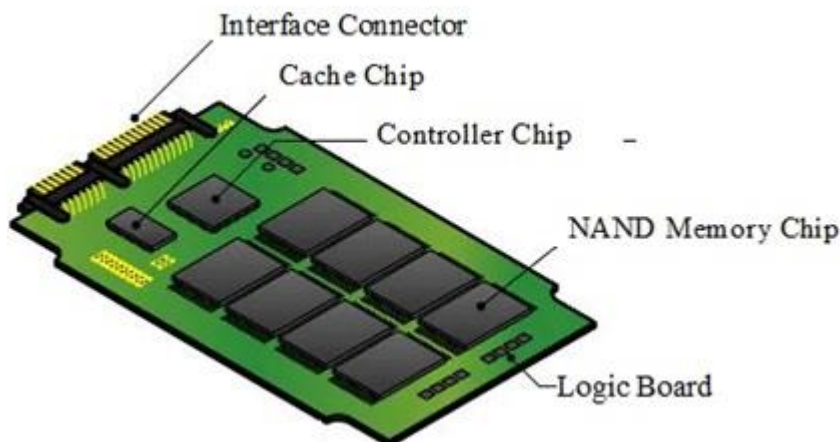
Unit 3: Computer - Solid State Drives (SSD)

3.0 Mian Contents

3.1 Computer - Solid State Drives (SSD)

3.1.1 What is Solid State Drive (SSD)?

A Solid State Drive (SSD) is a data storage device commonly employed in computers and different electronic devices. In contrast to conventional Hard Disc Drives (HDDs), which employ rotating discs for data retrieval and storage, Solid State Drives (SSDs) utilise NAND-based flash memory technology. Solid-state drives (SSDs) have experienced a surge in popularity owing to their different advantages which encompass enhanced speed, reduced energy usage, and heightened resilience.



Solid-state drives (SSDs) are capable of permanently storing data within an integrated circuit, commonly utilising flash memory technology. The utilisation of flash memory within an SSD facilitates the electrical and noiseless processes of data writing, transferring, and erasing.



Unlike mechanical hard disk drives (HDDs), SSDs do not have any moving components. Solid-state drives offer great performance and low noise levels due to their lack of moving components. However, it is important to note that SSDs are generally more expensive than HDDs.

In the past, SSDs possessed a more restricted storage capacity in comparison to conventional hard disc drives. However, presently, both SSDs and HDDs are available in a wide range of sizes to cater to diverse storage requirements. SSDs are frequently employed in premium computing systems or as supplementary storage components within personal computers intended for consumer use.

3.1.2 Characteristics of Solid State Drive

Some of the key Characteristics of Solid State Drive / Features of Solid State Drive are as follows –

- **Faster** – SSD devices are faster and enable systems to work instantly; it allows systems to boot immediately.
- **Multitasking** – The faster data access feature of SSDs supports the system to run multiple programs simultaneously.
- **Reliability** – SSDs are more reliable and perform well as they do not have moving parts that are susceptible to damage.
- **Maintains cooling** – SSD devices use flash memory which supports maintaining stable operating temperatures.
- **Flexible storage** – SSDs are available in a different form like mSATA which can be plugged into your system's motherboard and supports users to work along with your existing hard disc.

3.1.3 How does a Solid State Drive or SSD Work?

The working of Solid State Drive depends on its different key components. The description of Solid State Drive is as follows –

NAND Flash Memory

The primary component of an SSD is NAND flash memory. This is a type of non-volatile storage that retains data even when the power is turned off. NAND flash memory is made up of memory cells organized into pages and blocks.

Cells and Bits

Each memory cell in NAND flash can store multiple bits of data, typically one, two, or three bits per cell. The more bits a cell can store, the more cost-effective the SSD is, but it can also affect performance and longevity.

Pages and Blocks

Data is written and read from NAND flash memory in fixed-size pages, which are grouped into larger blocks. When data is written to an SSD, it is first programmed into an empty page. To update or delete data, SSDs use a process called wear-levelling, which ensures that data is evenly distributed across the NAND cells to prevent excessive wear on any one cell.

Controller

The SSD controller is a critical component that manages the reading and writing of data to and from the NAND flash memory. It handles tasks such as error correction, wear levelling, and garbage collection to optimize performance and longevity.

TRIM

The TRIM command is an important feature of SSDs. It allows the operating system to inform the SSD which data blocks are no longer in use, allowing the SSD to perform efficient garbage collection and wear levelling, which helps maintain performance over time.

Wear Leveling

NAND flash memory cells have a limited number of write-erase cycles before they degrade. Wear levelling ensures that data is written evenly across the memory cells, extending the lifespan of the SSD.

Read and Write Operations

SSDs can read data very quickly since there are no moving parts involved. Write operations, however, can be more complex because data must be written to empty pages and previously used pages must be erased before they can be rewritten. This process is managed by the SSD controller to optimize speed and minimize write amplification.

SSDs work by storing data in NAND flash memory cells, using a controller to manage read and write operations, and implementing various techniques like wear levelling and garbage collection to ensure longevity and maintain performance. Their speed, reliability, and efficiency have made them a preferred choice for storage in modern computers and electronic devices.

3.1.4 Advantages of Solid State Drive (SSD)

SSDs are more beneficial in different aspects; some of them are as follows –

- SSD devices have very low moving fractions which protect them from damage and run for a longer period.
- SSDs are substantially faster, with an immediate load of files to speed up the execution of programs.
- SSDs are lightweight.
- They have a long life.
- SSD storage capacity goes up and relative costs go down.

4.0 Pilot Questions

5.0 Summary

In this unit, you have learnt that:

6.0 Self-Assessment

7.0 Tutor Marked Assignment

8.0 References

MODULE 5: Computer - Memory Cards

Unit 1: Memory Cards

1.0 Introduction

3.0 Main Contents: Memory Cards

3.1.1 What are Memory Cards?

A memory card is a storage device used to store data and media files. It provides a permanent and non-volatile storage to store data. Memory cards are widely utilised in small, portable devices like cameras and phones.

Memory cards are a type of storage device that can hold data files. It provides space for the users to store data and files from the connected device. Memory cards act as electronic storage devices which are most widely used in cameras, camcorders, drones or mobile devices. Smartphones and digital cameras typically use SD and microSD memory cards. They are compatible with different devices, including DSLR and Nintendo Switch. However, not all cards are made equal; each device needs a different type of memory card.

3.1.2 Significance of Memory Cards

Memory cards are most widely used in digital devices and have its significance due to following reasons –

- **Storage Expansion** – They provide additional storage capacity.
- **Data Transfer** – Data can easily transfer between devices.
- **Portability** – Memory cards are smaller in size and have very low weight makes it portable.
- **Backup and Security** – They provide the way to keep important data for backup.
- **Versatility** – Memory cards are used in a broader sense, including digital cameras, cell phones, game consoles, and drones. Their adaptability makes them an important attachment for a wide range of digital devices.
- **Cost-Effective** – Adding a memory card is sometimes a more cost-effective solution than increasing a device's internal storage. This is especially useful for devices with little internal storage capacity.
- **High Performance** – High-speed memory cards can improve the performance of specific devices.
- **Durability** – Memory cards are designed to be durable.

3.1.3 How does a Memory Card Work?

Memory cards store data using flash memory technology. Flash memory is a type of non-volatile storage that can keep data long when the power is turned off. When you save data to a memory

card, it is written into the flash memory cells with electrical charges. These charges can be erased and rebuilt, allowing you to load new data onto the card.

3.1.4 How do You Choose SD Standard and Its Capacity?

Choosing a memory card entails determining what type of card your device needs, which is generally mentioned in the instruction manual. This usually indicates the SD standard for device compatibility. SD and microSD cards use the same standards: SD, SDHC, SDXC, and SDUC, respectively.

SDHC and SDXC are the two most popular current standards for both SD and microSD cards. The primary difference between SD standards is its storage capacity. If a user shoots 4K video, they most likely require an SDXC card, which has a maximum capacity of 2TB, more than enough for best video recording performance. SDXC cards also employ the exFAT file system to support big files created when recording video at high bitrates. In comparison, the FAT32 file system used by other card types limits file sizes to 4GB.

Type	Capacity
SD Card	2GB and under
SDHC Card	More than 2GB, up to 32GB
SDXC Card	More than 32GB, up to 2TB
SDUC Card	More than 2TB, up to 128TB

3.1.5 Types of Memory Cards

Some common types of memory cards are as –

1. Secure Digital (SD) Card

SD cards are one of the most popular forms of memory cards.



They are available with different data storage capacities, like MicroSD, miniSD, and normal SD is available. SD cards are widely used in digital cameras, smartphones, tablets, and other portable devices to increase storage space.

2. CompactFlash (CF) Card

A CompactFlash card is a memory card developed by SanDisk in 1994 that encompasses flash memory to store data on a compact portable device. It contains no moving mechanical elements and does not require a battery to save data.



SD cards are smaller and weaker than CompactFlash cards. Professional cameras and industrial applications use them.

3. Memory Stick

It is a removable flash memory card introduced by Sony in 1998. It was designed specifically to fix cameras, camcorders, portable music players, and computers to store and transmit data. Memory Stick cards were popular in Sony cameras, camcorders, and laptops.



Some common types of Memory Stick Cards are Memory Stick (Standard), Memory Stick PRO, Memory Stick Duo, Memory Stick PRO Duo, Memory Stick Micro (M2), AND Memory Stick XC.

4. XQD and CFexpress

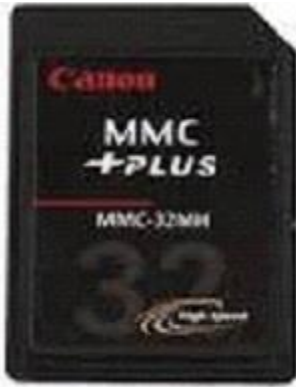
These are the high-speed memory cards used in professional cameras and video recorders. They can take high-resolution photographs and videos because of their fast read and write speeds.



These high-speed memory card types are used in professional cameras and video recorders. They can take high-resolution photographs and videos because of their fast read and write speeds.

5. MultiMediaCard (MMC)

MultiMediaCard (MMC) is a flash memory card that stores digital data. Siemens AG and SanDisk jointly introduced it in 1997. MMC cards are smaller than SD cards and used in older devices. Some common types of MultiMediaCard are RS-MMC (Reduced Size MMC), MMCplus, MMCmobile, SecureMMC, and eMMC (embedded MMC).



MMC cards are most widely used in digital cameras, mobile phones, portable media players, GPS devices, and personal computers.

6. SmartMedia Card

SmartMedia is a type of flash memory card developed by Toshiba in the mid-1990s. Most digital cameras and other portable devices no longer use SmartMedia cards because of their limited memory storage capacity.



Memory cards should be handled with care because they are prone to physical damage and data corruption. Regular backups of memory card data to external storage media can help to prevent data loss. Memory card storage capacity and data transfer rates vary. Memory cards should be compatible have adequate storage capacity, and be fast enough for their intended use. Choose a memory card that meets your specific requirements, such as high-speed video recording or data transfer.

Overall, a memory card is a type of storage media which is most preferably used to store different types of data, such as photos, videos, or other electronic information, within electronic devices. Commonly used devices that make use of a memory card encompass digital cameras, digital camcorders, handheld computers, MP3 players, PDAs, cell phones, game consoles, and printers.

3.1.6 Applications of Memory Cards

Some of the common applications of memory cards are as follows –

- **Computer system or Laptops** – Memory cards are used in computers and laptops to transfer the data.
- **Digital Cameras and Camcorders** – Used to store photos and Video Recording.
- **Mobile Phones and Tablets** – To support increasing storage capacity like storing photos, videos, and related data. Data can be easily transferred.

- **Drones** – Memory cards are used to capture aerial photos and videos
- **Media Players** – Memory cards are used to store music and video files to playback on media devices.
- **IoT Applications** – Memory cards are used to store sensor data and are compatible with industrial equipment for analysis.
- **Embedded Systems** – Memory cards are used to keep software and related applications in embedded systems.
- **Medical Devices** – Memory cards are used to keep medical records and patient information.

3.1.7 Advantages of Memory Cards

Some of the main advantages of memory cards are as follows –

- Memory cards are easy to fix in devices and use; user friendly.
- Memory cards are smaller in size and can be carried anywhere easily.
- Memory cards have comparatively low cost.
- Memory cards are compatible with almost all the devices where it is required to fix and data transformation can be successful.
- Allows fast data transfer speeds.

3.1.8 Disadvantages of Memory Cards

The disadvantages of memory cards are as follows –

- Memory cards are slow as compared to other systems' storage devices.
- Its data can be corrupt if proper ejection does not happen or if power failures.
- Memory cards can be stolen easily.
- Card readers are always required during data transfer which causes inconvenience.

4.0 Pilot Questions

5.0 Summary

In this unit, you have learnt that:

6.0 Self-Assessment

7.0 Tutor Marked Assignment

8.0 References

Unit 2: Computer - Optical Storage (Discs)

3.0 Main Contents

3.1 Computer - Optical Storage (Discs)

3.1.1 What are Optical Storage (Discs)?

Optical discs are a common method of storing huge amounts of data due to their large storage capacity and durability. Optical discs read and write data using laser beam. The optical storage technology has been widely employed to large quantity data like storing music, movies, games, and computer software.



Optical-storage technology uses a laser beam to imprint digital data onto an optical disc by creating small indentations in a spiral pattern on the disk's surface. A low-power laser scanner is employed to "interpret" these indentations, where changes in the brightness of reflected light from the indentations are transformed into electrical signals. This technology is used in to design compact disc, which is capable of recording sound; the CD-ROM, which can store text, images, and sound; the WORM (write-once read-many) disc, which allows for one-time writing and multiple readings; and newer discs that offer complete rewritability. Overall, they are flat, circular polycarbonate discs with a reflective data layer and a protective outer layer.

3.1.2 Characteristics of Optical Storage Systems

Some of the key Characteristics or features and properties of Optical Storage Systems are as follows –

- **High Storage Capacity** – Optical storage has large storage capacity; CDs, DVDs, and Blu-ray discs has high storage capacity.
- **Performance** – The performance of Optical storage devices are up to mark as they deliver fast data transfer rates, less access time and high-speed buffering.
- **Efficient reading and writing of data** – Optical storage has efficient reading and writing of data.
- **Durability and Reliability** – The architectural design of optical storage devices reduces risk of damage or data corruption due to surface scratches, dust, and fingerprints. These features make it durable and reliable.
- **Error correction** – Optical media reading technology uses error correcting techniques like Reed-Solomon and Cross-Interleaved Reed-Solomon Error correcting to ensure data integrity.

3.1.3 Types of Optical Storage

Some of the common types of Optical Storage are as follows –

1. CD (Compact Disc)

The Optical Storage CDs were introduced in the early 1980s. In optical storage technology; CDs were introduced first. They can hold up to 700 megabytes of data and are commonly used for music, software, and data storage.

2. DVD (Digital Versatile Disc)

DVDs are extended version of CDs; they can have significantly more storage capacity like 4.7 GB single-layer and 8.5 GB dual-layers. This optical storage is most widely used in movie production, software distribution, and data backup.

3. Blu-ray Disc

Blu-ray is an advance optical discs storage method. A Blu-ray, may hold more data than DVDs. Standard discs can carry 25 GB (single-layer) or 50 GB (dual-layer), with larger capacities for specialised variants. Blu-ray discs are used for both HD video and data storage.

4. Archival Disc

Archival optical discs keep data for a long time. They are ideal for archiving data due to their large volume and longevity.

3.1.4 How Optical Discs work?

Optical discs work by using a laser to read and write data. Its working depends on the structure of an optical disc. The layered architecture includes –

- **Label** – A layer that can contain either printed text or images.
- **Polycarbonate layer** – A transparent overlay containing the actual data.
- **Reflective layer** – A thin coating of material that reflects laser light back onto the laser diode.
- **Protective layer** – A clear layer that shields the reflective layer from damage.
- **Recording Mechanism** – Lasers leave tiny pits and lands on CDs, DVDs, and Blu-rays. Pits and lands represent binary data.
- **Reading Mechanism** – The laser of the disc drive detects the reflections of pits and lands in order to read data. Drive translates reflections into digital data.

A less powerful laser is used to extract digital data from the designs on the reflecting layer. To write data, a more powerful laser burns a small portion of the reflecting layer, creating pits and bumps that represent binary data. Optical discs, a common form of data storage and video distribution, are increasingly competing with other storage technologies such as USB drives, external hard drives, and cloud storage. However, they are still employed in some cases where longevity, the capacity to keep data even when the machine is turned off, and resistance to data corruption are critical.

3.1.5 Advantages of Optical Storage

Some of the key advantages of Optical Storage are as follows –

- **Durability** – The lifespan of optical storage media is quite long which makes its durable.
- **Environmental resistance** – Optical discs are resistant to certain environmental variables, such as humidity and temperature. This increases their durability and stability.
- **Error correction** – Optical media reading technology uses error correcting techniques to correct errors to ensure data integrity.
- **High storage capacity** – The storage capacity of optical storage media is higher.
- **Strong performance** – The performance of Optical storage devices are up to mark as they deliver fast data transfer rates, less access time and high-speed buffering.
- **Reliability** – The Optical storage devices are reliable.

3.1.6 Disadvantages of Optical Storage

Some of the key disadvantages of Optical Storage are as follows –

- **Compatibility** – the architectural design of Optical Storage includes different disc formats and standards which makes compatibility issues.

- **Access time** – Generally, optical storage media has slower access times.
- **Scratch and dust sensitivity** – optical discs are susceptible to scratches and dust.
- **Disk capacity** – Disk storage capacity can be improved.
- **Plastic disks** – The plastic material used to frame an optical disk is not eco-friendly.

Unit 2: Computer - NAND Flash Memory

1.0 Introduction

2.0

3.0 Main Contents: Computer - NAND Flash Memory

3.1 What is NAND Flash Memory?

NAND flash memory is a non-volatile memory which doesn't require a regular power supply to work and retain data. Its main goal is to read and write data from and to the memory very fast and maximise its storage capacity to store data large-size data and reduce cost.

NAND flash memory stores data into memory blocks, when the power supply gets off the metal-oxide semiconductor activates charges to keep memory cells functional. NAND memory cells are made up of two types of gates: control and floating gates. Both gates are used to regulate the flow of data. To program one cell, a voltage charge is applied to the control gate. Generally, NAND flash memory has digital cameras, MP3 players, USB flash drives, smartphones, tablets and laptops.

3.1.2 Where NAND Flash Memory is Used?

NAND is commonly found in memory cards, USB flash drives, and solid-state drives. NAND Flash is most widely used in cell phones, laptops and tablets in our homes and offices. It is also used in traffic lights, digital advertising panels, passenger announcement systems, display devices or any other artificial intelligence (AI) application where data transfer and retain data is likely to include NAND Flash.

NAND flash memory is evolving to meet the demands of modern data storage. Its robustness, speed, and cost-effectiveness make it a key player in the ever-changing world of memory storage technology.

3.1.3 Why NAND Flash Memory is Important?

When compared to traditional hard disk drives, NAND Flash-based devices provide shock resistance, great endurance, and the ability to handle extreme pressure and temperature. They have no moving parts, unlike HDDs, and are therefore suitable for systems that make vibrations and shake. NAND also provides rapid read access times, which are critical when processing large amounts of data. Overall, NAND is best suited for applications that require large amounts of data storage. It provides higher densities, larger capacity, and reduced costs. It does speedier erases, sequential writes, and sequential reads.

- **Most widely used in Consumer Electronics** – It is an essential component in a wide range of electronic gadgets, including smartphones, tablets, digital cameras, and portable media players to store data.
- **Critical for Solid-State Drives (SSDs)** – SSDs, which use NAND Flash Memory, provide considerable performance over traditional hard drives.
- **Portability** – Its high storage density enables the development of lightweight storage devices such as USB flash drives and memory cards, which are required for portable data storage and transfer.

- **Power Efficiency** – NAND Flash Memory requires less power than mechanical hard drives, which is critical for battery-powered devices like laptops, smartphones, and tablets. This efficiency increases battery life and improves the user experience.
- **Reliability and Durability** – NAND Flash Memory, unlike traditional hard drives, is more resistant to physical shock and damage since it contains no moving parts. This makes it perfect for transportable and hard applications that require high reliability.
- **Scalability and Cost-Effectiveness** – The technology continues to advance, providing larger storage capacities at lower costs per gigabyte. This allows for the cost-effective storage of large amounts of data, which provides benefits for both consumers and enterprise applications.

3.1.4 Features of NAND Flash Memory

Some of the key characteristics of NAND Flash Memory are as follows –

- **Non-Volatile Memory** – NAND flash memory is non-volatile, which means that it retains recorded data without the use of a battery or other power source which makes it suitable for long-term data storage. Such memories can preserve data for a long period without being damaged.
- **High Density** – NAND flash memory is available with high storage capacity in a small physical space. Its high-density feature makes it usable in SSDs, USB drives, and memory cards.
- **Durability** – It can write and erase data a number of times which makes it durable.
- **Performance** – NAND flash memory allows fast read and write operations.
- **Cost-Effectiveness** – NAND flash memory is cheaper per gigabyte as compared to other types of memory.
- **Reprogrammable** – NAND flash memories are that they can be programmed to meet the user's needs. The programmable IC chips may store any type of data, from documents to films and allow users to use them whenever required.
- **Faster Write and Erase Time** – NAND flash memory is fast read and writes data; due to this feature, it is most widely used in the application areas where data writing operations are required. It is also a distinguishing property of NAND flash memory that takes less time to erase its contents.
- **More Storage** – NAND flash memory can store up to 2TB storage devices. Modern NAND flash memories are launching to store large-size data.
- **Low Manufacturing Cost** – Its manufacturing cost is comparatively low.
- **Data Integrity and Reliability** – Advanced error-correcting codes (ECC) and wear levelling algorithms are used to improve data integrity and extend memory life.
- **Scalability** – NAND Flash can be scaled to extremely high capacity, which is a driving factor for large-scale data storage solutions.

3.1.5 How does NAND Flash Memory Work?

NAND Flash Memory operates through a series of processes that involve the storage and retrieval of data using floating-gate transistors. Its overall working depends on its structure, data storage functionalities, and reading and erasing data. A brief description of the different phases is as follows –

3.1.6 Structure

NAND Flash is made up of memory cells that store bits of data. Each cell is made up of a floating-gate transistor with an additional gate (the floating gate) between the control gate and substrate. These cells in memory are divided into pages, the smallest writing units. Pages are divided into blocks, the smallest erasable components.

Storing Data

The floating gate can store an electrical charge that represents data; the presence or absence of this charge marks whether the cell stores the binary '0' or '1'. To write data, apply a high voltage to the control gate, which causes electrons to tunnel through the thin oxide layer and become trapped on the floating gate. The threshold voltage of the transistor varies according to the charge stored on the floating gate, allowing the cell to represent different states.

Reading Data

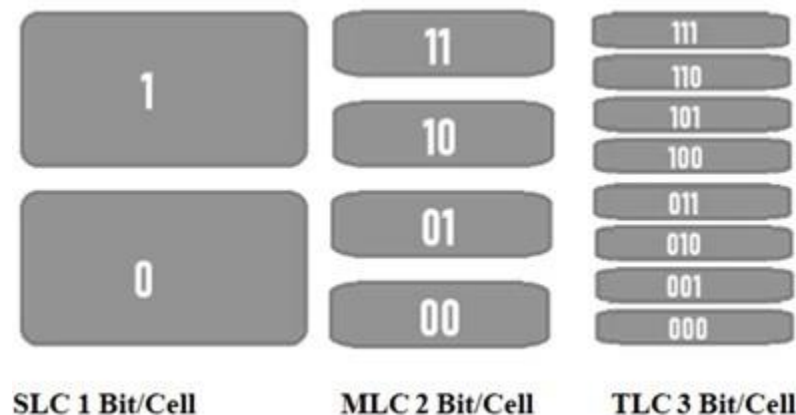
To read data, a voltage is delivered to the control gate. The amount of current flowing through the transistor is determined by the charge stored in the floating gate. The sensor circuit detects current flow and identifies the stored data ('0' or '1') based on the transistor's threshold voltage.

Erasing Data

NAND Flash memory cannot delete individual bits or pages; instead, it erases entire blocks. To erase a block, a high voltage is supplied in the opposite direction, removing electrons from the floating gate and restoring the cells to their default state (typically '1').

3.1.7 Types of NAND Flash Memory

Common types of NAND flash storage include SLC, MLC, TLC, QLC and 3D NAND. A brief description of these is as follows –



SLCs

SLC is an abbreviation for single-level cells. It is one of the older NAND Flash, with each cell representing only one binary digit (bit). SLCs, or single-level cells, hold one bit per cell. SLC offers the best endurance, but it is also the most expensive type of NAND flash storage.

Multi-Level Cell

MLC, or multi-level cells, store two bits per cell; hence each Multi-Level Cell may hold two bits. These cells have less endurance than SLCs since Write Cycles and ensure occur twice as frequently as SLCs.

Triple-Level Cell

TLC stands for Triple-Level Cell. As the name implies, each cell contains three bits of memory, allowing for greater data storage in the same footprint. TLC drives are primarily used by enterprise and consumer companies.

Quad-Level cell

Each Quad-Level cell can represent up to four bits because the name "Quad" signifies "four". QLC memory is the technology that offers the lowest per-terabyte cost when compared to hard drives.

3D NAND Flash Memory

3D NAND flash saves space by building memory cells vertically in a three-dimensional matrix. This method of producing a NAND Memory storage chip saves a lot of physical space, making the chip smaller.

4.0 Pilot Questions

5.0 Summary

In this unit, you have learnt that:

6.0 Self-Assessment

7.0 Tutor Marked Assignment

8.0 References

Unit 3: Computer - NVMe Memory

1.0 Introduction

3.0 Main Contents

3.1 What is Non-volatile Memory Express (NVMe)?

NVMe (Non-Volatile Memory Express) is a protocol that uses the PCI Express (PCIe) interface to link solid-state drive (SSD) storage to servers or CPUs. It is the fastest and most popular protocol for connecting to a computer's motherboard's non-volatile memory express (NVMe). NVMe was developed in 2011 by a group of IT professionals; this device is an alternative to SATA and Serial Attached SCSI (SAS) protocols which provides higher throughput for faster and more reliable performance. In addition to storage and technology advancements, NVMe helped to accelerate the development of critical technologies like as the Internet of Things (IoT), artificial intelligence (AI), and machine learning (ML).

The NVMe protocol is dedicated to data transfer with low system overheads per input/output operation per second. It is used in SSDs with flash memory. NVMe addresses the bottlenecks in the SATA interface and communication protocols. NVMe technology uses the PCIe bus instead of the SATA bus to unleash massive bandwidth potential for storage devices. NVMe technology unlocks different possibilities for storage devices by increasing economy, performance, and compatibility across different platforms.

Recent technical developments in data storage have pushed organizations and individuals to move from traditional hard disk drives (HDDs) to fast data transmission, and lower-latency solid-state

drives (SSDs). SSD and NVMe are two different types of drives that can be used to complement each other. SSDs are semiconductor-based storage that includes flash storage, and NVMe is a data transfer protocol that uses reduced system overheads per input/output operation per second in SSDs with flash memory. It is expected that the NVMe technology will become the new industry standard in the upcoming time.

3.1.1 Why NVMe is Important?

NVMe has a Peripheral Component Interconnect Express (PCIe) bus which can achieve transfer speeds of up to 20 gigabytes per second (Gbps). Power is not required for the NVMe device to retain its memory. Unlike other technologies, NVMe storage can connect directly to a computer's CPU using a PCIe socket and enable the drive's flash memory to function through PCIe.

NVMe technology has fast data transfer speeds and allows for parallelism and polling; NVMe SSDs can optimize response times, and write speeds, and reduce latency to avoid CPU bottlenecks. Additionally, NVMe technology uses less power than the popular Small Computer System Interface (SCSI).

3.1.2 Characteristics of NVMe

Some of the key characteristics of NVMe are as follows –

- **High Throughput and Low Latency** – NVMe leverages the high-speed PCIe bus for latency which makes faster data transfer rates.
- **Performance Enhancement** – Its low latency and high-speed data transfer features significantly improve the performance of storage systems.
- **Scalability** – The structural design of NVMe makes it scalable to improve performance by providing compatibility with future storage advancements.
- **Cost Efficiency** – Its higher performance and durability typically result in long-term cost advantages due to increased productivity and decreased downtime.
- **Future-proofing** – NVMe is highly extensible and compatible with all emerging persistent memory technologies.

3.1.2 How does NVMe Work?

NVMe is a storage transfer technology that allows you to quickly access data from flash memory storage devices like SSDs. It allows flash memory to communicate directly with a computer using a high-speed peripheral component interconnect express (PCIe) bus, resulting in quick, high-throughput, and parallel data transfer. NVMe communicates between a storage interface and a system CPU via a high-speed Peripheral Component Interconnect Express (PCIe) bus. The NVMe protocol was developed to work with fast media.

NVMe's work starts with the host; they write I/O commands. The NVMe controller retrieves and executes I/O command queues, then returns I/O completion queues, followed by an interrupt to the host. The host logs I/O completion queues and clears the door register. It reduces overhead as compared to standard transfer protocols such as serial attached SCSI (SAS) and serial ATA (SATA), NVMe is optimized for non-uniform memory access (NUMA), which means it can handle queues across multiple CPU cores.

3.1.3 Advantages of NVMe

Some of the key advantages of NVMe are as follows –

- NVMe technology gives storage, speed and compatibility features.
- It uses PCIe sockets which makes faster data transmission.

- NVMe communicates directly with the system.
- NVMe drives work with all major operating systems regardless of form factor.
- It is most suitable to utilise SSDs and communicate between the storage interface and the system's CPU using PCIe.
- Its performance is significantly higher.
- NVMe is specifically intended for SSDs and will eventually become the next industry standard.

MODULE 6: COMPUTER - COMPACTFLASH CARD

Unit 1: Computer - CompactFlash Card

3.0 Main Contents CompactFlash Card

3.1.1 What is a CompactFlash Card?

A CompactFlash card is a memory card used to store data. It is also known as CF card which was developed by SanDisk in the year 1994. A CompactFlash card is a type of flash memory that stores the data on a tiny chip. This card does not have moving mechanical parts; so, it does not require a battery to retain data.



A CompactFlash card is one of the most widely used portable storage devices which is smaller in size and a more reliable storage device. It is most widely used in computing devices to provide faster and reliable data storage. It is a type of flash memory used as a removable storage for digital images and video cameras, but it can also be found on personal devices such as PDAs and portable music players.

A CompactFlash card was initially offered with storage capacities ranging from megabytes to several gigabytes. Modern CF cards can offer storage capacities of up to 512GB or more. The earliest CompactFlash cards were designed using NOR flash memory that can read and write data in random order. This feature made it ideal and popular to store images or text-type data.

NOR flash memory was slower and more expensive; later on, flash memory used NAND flash memory and became more popular and less expensive. NAND flash memory is speedier and can be written in serial order, making it ideal for storing large files like video or music.

3.1.2 Significances of CompactFlash Card

CompactFlash cards are available with different storage capacities which enable it for substantial storing of high-resolution photos, films, and other significant files. This is especially useful for photographers and videographers who need to save a lot of data. These cards are available with different storage capacities ranging from 8GB to 512 GB. A CompactFlash card's capacity is defined by the number of NAND flash chips used for data storage. CompactFlash cards are more reliable in harsh environments. Nowadays, CompactFlash cards are used in industrial applications and professional use in the field.

CompactFlash cards are widely used and still compatible with many high-end cameras, camcorders, and other professional equipment. Their extensive history in the industry provides

continuing maintenance and compatibility with legacy equipment. CompactFlash cards offer consistent and stable performance, which is crucial for applications requiring uninterrupted data storage and retrieval. CompactFlash cards are intended to protect data integrity, lowering the chance of loss or corruption. This is vital for professionals who need to protect their work, as well as for industrial applications that require data integrity.

Despite the development of newer memory cards like SD cards and microSD cards, CompactFlash cards remain useful in situations where their longevity, high capacity, and consistent performance, are necessary.

3.1.3 How does CompactFlash Card work?

CompactFlash cards work by plugging them into pinholes for the card to function correctly. CF cards are easy to use. They use FAT32 or exFAT formatting and are compatible.

CompactFlash cards employ NAND flash memory. Data is stored in memory cells that are grouped into blocks. Each cell contains a set number of bits, usually one bit for single-level cells (SLC) and more for multi-level cells (MLC). Traditional CF cards use the Parallel ATA (PATA) interface for transferring data between the card and the host device while advanced CompactFlash cards may employ the CFexpress standard, which uses the PCIe interface for higher data transfer rates.

Each CF card comprises a controller for data storage and retrieval. The controller is in charge of functions such as wear levelling and error correction. When data is written to the card, the controller assigns it to the proper memory cells. During a read operation, the controller receives data from the cells and transfers it to the host device. CF cards require a power supply to function. When plugged into a device, the device supplies the electricity required for the card to work.

3.1.4 Types of CompactFlash Cards

CompactFlash (CF) cards are come in different types; these are as follows –

CompactFlash (CF) cards Type	Description
CF Type I	Its a standard CompactFlash card measuring 3.3mm in thickness. It is most widely used in digital cameras, camcorders, and other related devices.
CF Type II	These are thicker at 5mm; their storage capacity is comparatively more.
Cfast	These cards use a Serial ATA (SATA) interface to work on faster data transfer principles. It has a 3.3mm thickness. It is used in cameras, camcorders, and other professional equipment.
Cfexpress	These cards use the PCIe and NVMe interfaces which have higher data transfer rates. Its thickness may vary in size. These cards are more likely used in video recording and burst photography.

UDMA (Ultra Direct Memory Access) CF Cards	These CompactFlash cards use UDMA protocol and have higher data transfer rates. It has a 3.3mm thickness. These are most widely used in professional photography and video applications.
Industrial CF Cards	These CF cards are designed to survive high temperatures, vibrations, and shocks. Its thickness may vary in size. These are most widely used in industrial PCs and automation systems.

3.1.5 Features of CompactFlash Card

Some of the key characteristics of CompactFlash Card are as follows –

- **Higher Performance** – CompactFlash cards are much faster than other memory cards, such as SD cards.
- **Capacity** – CompactFlash cards are available with different storage capacities ranging from 8GB to 512GB. This makes them perfect for storing huge files like high-resolution photos or films.
- **Durability** – CompactFlash cards are robust and durable. They include no moving parts and are housed in a solid plastic housing.
- **Reliability** – CompactFlash cards are extremely reliable and have a low failure rate. They include features like Non-Volatile Memory, Error Correction and Wear Leveling. They do not have any moving parts that can fail or wear out; this feature makes them more reliable.
- **Compatibility** – CompactFlash cards are very reliable and have a low failure rate. They have no moving parts that could break or wear out.
- **Power Consumption** – CompactFlash cards are designed to be power-efficient, reducing the strain on the devices battery.
- **Security Features** – CompactFlash cards have built-in security features like password protection and encryption to keep them safe from unauthorised users or unauthorised access.

3.1.6 Applications of CompactFlash Card

CompactFlash cards are most widely used in different domains of the industry. Some of the most commonly used application areas are as –

- CompactFlash cards are most widely used in digital cameras and camcorders.
- CompactFlash cards are used for video recording.
- CompactFlash cards are used in portable music players and personal digital assistants (PDAs).
- CompactFlash cards are used in data logging and machine automation.
- CompactFlash cards are used in healthcare to store patient data and images.
- CompactFlash cards are also used to back up important data as a Data Backup device.
- Because of their endurance and capacity to tolerate severe temperatures, CF cards are employed in aerospace and military applications for data storage and transfer.
- CompactFlash cards are used in entertainment systems also to store media files.
- CompactFlash cards are used in scientific research to store research data.
- CompactFlash cards are also used as environmental monitoring devices to store data collected from various sensors.

UNIT 2:

1.0 Introduction

2.0

3.0 Main Contents: What is Cloud Storage & How Does It Work?

3.1 What is Cloud Storage?

Cloud storage is an advanced mechanism of data storage and its processing on remote servers that are managed and administered by cloud engineers or cloud experts. Cloud services are provided by cloud service providers that do not directly interact with users. Typically, the administration of cloud data and related activities occurs using internet connectivity as compared to data storage on hard drives or on-site servers. Cloud storage services are more beneficial in terms of their scalability, accessibility, cost-effectiveness, and data redundancy.

Users can access Cloud Storage data using an internet connection and software such as a web portal, browser, or mobile app via an application programming interface (API).



Data storage stores data for long-term use, while in a common sense RAM stores data and instructions for rapid CPU execution. Overall, the system requires storage for capacity and memory for speed and responsiveness accordingly.

Cloud Storage saves data on remote servers, including files, corporate data, movies, and photographs. Users upload data to servers using an active internet connection, which is then saved on a virtual computer on a physical server. To ensure availability and redundancy, cloud companies frequently distribute data over different virtual computers in data centres situated across the world. If storage requirements increase, the cloud provider will deploy additional virtual computers to manage the data load.

During selecting a cloud storage provider; an individual or business professionals should examine their specific storage needs, price, data security, and the service provider's market value, trust and business protocols. Different cloud service providers may have varying features and pricing structures, so choose one that meets your specific needs.

1.2 Why Cloud Storage is Important?

Cloud Storage allows organisations to store, access, and preserve data without the need to own and operate their own data centres, reducing costs from capital expenditure to operation. Cloud storage is scalable, allowing organisations to extend or reduce their data footprint as needed. Cloud storage is typically less expensive than on-premises storage. This is one of the primary benefits of cloud data storage. Furthermore, cloud data storage provides scalability for your company as its data storage requirements grow. This also relies on the service provider you use to store your data. Cloud data storage offers a variety of storage possibilities. Cloud data storage allows for the storage and transport of a wide range of business data, including sensitive information. Cloud data storage allows you to access your data from any device with an internet connection. This is especially important for organisations that wish to give their staff the flexibility to work remotely.

1.3 How does Cloud Storage Work?

Cloud Storage uses remote servers to save data, such as files, business data, videos, or images. Users upload data to servers via an internet connection, where it is saved on a virtual machine on a physical server. To maintain availability and provide redundancy, cloud providers will often spread data to multiple virtual machines in data centres located across the world. If storage needs increase, the cloud provider will spin up more virtual machines to handle the load. Users can access data in Cloud Storage through an internet connection and software such as a web portal, browser, or mobile app via an application programming interface (API).

1.4 Types of Cloud Deployment Models

When implementing cloud architecture, there are three types of cloud deployment models available to deliver cloud services: public cloud, private cloud, and hybrid cloud. Hence, cloud storage is available in private, public and hybrid clouds.

1.5 Public Storage Clouds

In this model, a user can connect using the internet to a storage cloud managed by a cloud provider and shared by other businesses. Providers often make services available from almost any device, including cell phones and PCs, and allow users to scale up and down as needed. Compute, storage, networking, develop-and-deploy environments, and applications are all delivered through the internet by public clouds. Public clouds are owned and operated by third-party cloud service providers such as Google Cloud.

Unit 3: Private Cloud Storage

1.0 Private Cloud Storage

Private clouds are created, managed, and used by a single organisation and are often on-premises. They offer better control, customisation, and data security, with similar costs and resource restrictions as traditional IT environments. Private cloud storage configurations often imitate the cloud paradigm, but they are located within your network, utilising a physical server to generate instances of virtual servers to expand capacity.

iii. A user can opt to take complete control of an on-premise private cloud or hire a cloud storage provider to create a dedicated private cloud that can be accessed using a private network. Banks and retail companies may prefer private cloud storage due to the privacy of the data they handle and keep.

1.1 Hybrid Cloud Storage

A Hybrid model incorporates characteristics of both private and public clouds, allowing organisations to choose which data to store in which cloud. For example, highly regulated data with secure archiving and replication requirements is typically better suited to a private cloud environment, but less sensitive data (such as email that does not include company secrets) can be hosted in a public cloud. Some businesses utilise hybrid clouds to augment their internal storage networks with public cloud storage.

1.2 Types of Cloud Storage Services

Some common cloud storage services are as follows –

Software as a Service (SaaS)

This service is used for business applications. As its name implies, Service providers provide software such as 'Software as a Service (SaaS) to business professionals to use like Marketing and sales automation, human resource software, and financial management software. The software programs or applications are hosted on the cloud and accessed using the internet by users.

Platform as a Service (PaaS)

PaaS is a cloud service that provides all of the tools needed to develop and run an application. This comprises a programming language, a code editor, a runtime, a database, a file system, and an execution environment.

Infrastructure as a Service (IaaS)

IaaS is a cloud service that offers the infrastructure required to operate an application. This encompasses virtual machines, networks, storage, and load balancers.

Cloud Storage Providers

There are many Cloud Storage Providers are exist in the market. Some of the trusted leaders of Cloud Storage Providers are as follows –

Google

Google is one of the most popular cloud storage service providers that provides Google Drive as a Cloud Storage. Google Drive is most popular amongst users; Google Drive gives 15 GB of free cloud storage to its users. Users like Google Drive because of its simplicity and ease of use.

Amazon

Amazon offers free cloud storage with 5 GB of free space for its users. It also allows you to gain additional free Amazon Drive space. Participating in Amazon Drive's referral programme, a user can earn up to 50 GB of free space.

Microsoft

Microsoft OneDrive offers 15 GB of free online storage. You can recommend friends and get an extra 5 GB of space.

1.3 Characteristics of Cloud Storage

Some of the key features or characteristics of cloud storage are as follows –

Accessibility

If you have access to the internet, you can access data stored on the cloud from anywhere. As a result, it is much easier for individuals and businesses to access their data using multiple devices like personal computers, cell phones, and tablets.

Cost-effectiveness

Cloud storage typically charges users for the storage capacity and resources that they use. This may be less expensive than updating on-premises storage.

No Upfront Cost

A user does not spend money to implement an effective backup.

Security

Cloud service providers prioritise the security of their users' data. This includes access limits, security assessments, and continuous data encryption and decryption. Users must be responsible for the security and setup of their data in the cloud.

Scalability

Because of the cloud's high scalability, users can modify the amount of data stored there to fit their needs. This adaptability is helpful for businesses with variable storage requirements.

Backup and Disaster Recovery

Cloud storage services include backup and restore tools. Users can schedule frequent backups and retrieve data if they lose data or the system fails.

1.4 Advantages of Cloud Storage

Cloud storage is beneficial in different aspects; some of the advantages are as follows –

- **Usability and accessibility** – Cloud data storage services are available with an easy-to-use user interface and include drag-and-drop features.
- **Security** – Generally, cloud storage providers keep copies of your data at different data centres, so even if the data gets lost or corrupted at the server, the backup will be there.
- **Cost Effective** – Cloud data storage can help to reduce internal expenses.
- **Convenience** – Cloud storage can be accessed from any device.
- **Sharing of files** – All cloud storage options provide file-sharing features, which help you to share your files with other users.
- **Automation** – Cloud storage services work like a hard disk on your system. If you want to store a file in the cloud, it will not affect ongoing tasks.
- **Multiuser support** – With cloud storage, multiple users can collaborate with a common file.
- **Scalable** – Cloud storage is scalable; if your current plan of storage is not enough, you can upgrade the service plan. In this case, a user does not need to move any data from one location to another.
- **Disaster recovery** – Data backup features are available on cloud storage; in case a disaster occurs, users do not need to worry because their data can be utilised using data backup.

1.5 Disadvantages of Cloud Storage

Below are the main drawbacks of using cloud storage services –

- **Internet Dependency** – Without an internet connection, you will be unable to access your data while downloading a file from cloud storage.
- **Data Security and Privacy** – Many cloud storage companies fall short in terms of data security and privacy, and there are some incidents of cloud storage data breaches.

- **Costs** – Most of the top cloud storage services are costly since they are primarily intended for commercial use.

MODULE 7: COMPUTER - MEMORY

Unit 1: Computer - Memory

What is Computer Memory?

A physical device that stores data or information temporarily or permanently in it is called memory. It is a device where data is stored and processed. In common, a computer has primary and secondary memories. Auxiliary (secondary) memory stores data and programs for long-term storage or until the time a user wants to keep them in memory, while main memory stores instructions and data during programme execution; hence, any programme or file that is currently running or executing on a computer is stored in primary memory.

Types of Computer Memory

Computer memory comes in various types and serves different purposes –

- **Primary Memory (RAM - Random Access Memory)** – Volatile memory loses its contents when the machine is turned off. RAM stores the data that is actively being used. During the booting process of a system, the operating system actively uses RAM and applications that are necessary to execute a file or a program. It speeds up CPU processing by providing fast data and instruction access.
- **Secondary Memory (Storage)** – Secondary Memory is also known as permanent memory or non-volatile memory of a computer. Secondary memory retains data when the machine shuts down. Files, programmes, and the OS are stored there permanently. HDDs, SSDs, USB flash drives, and optical discs are non-volatile memory devices.
- **Cache Memory** – Memory that is smaller and faster than RAM is called cache memory. It is placed closer to the CPU than the RAM. It saves data and instructions that are used a lot so that processing goes faster.
- **Register Memory** – Register memory, which is also called processor registers or "registers," is the smallest and fastest type of computer memory that is directly integrated into the CPU. Registers are small, fast storage units inside the CPU that are used to quickly store data that is being processed or instructions that are being run.

Primary Memory

Computer primary memory is also known as volatile memory, as it loses contents when the machine is turned off. There are two types of primary memory: RAM and ROM. RAM stores the data that is actively being used. During the booting process of a system, the operating system actively uses RAM and applications that are necessary to execute a file or a program. It speeds up CPU processing by providing rapid data and instruction access.

Here is the classification of primary memory:

- **RAM (Random Access Memory)**
 - SRAM (Static RAM)
 - DRAM (Dynamic RAM)
 - SDRAM (Synchronous DRAM)
 - DDR (Double Data Rate) RAM
- **ROM (Read-Only Memory)**
 - PROM (Programmable ROM)
 - EPROM (Erasable Programmable ROM)
 - EEPROM (Electrically Erasable Programmable ROM)
 - Flash ROM

1. Random Access Memory (RAM)

RAM contains of a various number of memory locations wherein each location, typically 8-bits are stored. It can be possible to read from a RAM location, as well as write to a RAM location. The drawback of RAM is that it is volatile. That means, when the power supply to the RAM is switched off, the information in the RAM will be lost.

From the memory, data can be accessed in two different ways – Sequential Access and Random Access.

Sequential Access – Sequential access and random access are two types of accessing information. In sequential access, it is mandatory to access information strictly in order. If there are 4000 memory locations, it have to be accessed in the order of 1, 2, 3,...,4000. Thus, it takes minimum time to access information from location 0 and at most time to access information from location 4000. Magnetic tape is an example that employs sequential access.

Random Access – In a random access technique, it can be possible to access a memory location in any order. For example, one can read from the 4000 locations in the order of 1500, 1210, 3060, 1640, 1352, and so on. Second, it takes a similar time to read from a memory location irrespective of its position. In a RAM, the access method is random, and its name, in fact, is derived based on the method of access.

2. Read Only Memory (ROM)

ROM consists of a number of memory locations wherein each location typically stores 8 bits like a RAM. A ROM also uses random access method just like a RAM. The advantage of ROM is that it is non-volatile in nature. That means, though the power supply to the ROM is switched off, the information in the ROM will not be lost. However, only reading operation is possible from a ROM location. Thus, in a computer, ROM is used for storing information which is not lost when power is switched off.

Let us discuss different versions of ROM provided by semiconductor manufacturers for storing information:

Mask-Programmed ROM It derives this name because the information is written to this type of ROM at the time of manufacture by applying a suitable mask. Once written, the information cannot be changed, even by the manufacturer. It is cheaper compared to other types of ROMs when considering cost per unit quantity, but requires bulk ordering. It is used in equipment produced in large quantities.

Programmable Read Only Memory (PROM) The user writes information to this type of ROM using a PROM programmer. Once written, the information cannot be changed. Like Mask-Programmed ROM, the information is permanent. It is more expensive than mask ROM but allows purchasing in smaller quantities.

Erasable Programmable Read Only Memory (EPROM) Information is written by the user with an EPROM programmer. As its content is erasable and rewritable, the user can modify it multiple times. Data is erased using strong ultraviolet (UV) light on the quartz window of the EPROM chip, which removes all content. Users can purchase a single piece of EPROM and rewrite its content multiple times.

Electrically Erasable Programmable Read Only Memory (EEPROM) Information is written by the user using an EPROM programmer. Unlike EPROM, EEPROM data is erased using electrical signals rather than UV light. EEPROM allows selective data erasure and is more expensive than other ROM types. It is gaining popularity due to its flexibility.

Types of ROM	Info.entered by	Possible to change info?	Cost per chip	Min.order quantity	Method of erasing info.	Application area
Mask-programmed ROM	Manufacturer	No	Cheapest	Large	Not possible	Large-scale production
PROM	User	No	Costlier	One	Not possible	Medium-scale production
EPROM	User	Yes	Even more costly	One	Using strong UV source	R& D
EEPROM	User	Yes	Costliest	One	Using electrical signals	R& D

Secondary Memory

Computer secondary memory stores data and programs permanently, even when the computer is off. The secondary memory is also known as external memory or auxiliary memory.

Classification of Secondary Memory

Here is the complete classification of secondary memory:

- **Magnetic Storage**
 - Hard Disk Drive (HDD)
 - Floppy Disk (Obsolete)
 - Magnetic Tape
- **Optical Storage**
 - CD (Compact Disc)
 - CD-ROM (Read-Only Memory)
 - CD-R (Recordable)
 - CD-RW (Rewritable)
 - DVD (Digital Versatile Disc)
 - DVD-ROM
 - DVD-R/DVD+R
 - DVD-RW/DVD+RW
 - Blu-ray Disc (BD)
- **Flash Storage (Solid-State Storage)**
 - Solid-State Drive (SSD)
 - USB Flash Drive (Pen Drive)
 - Memory Cards (SD Card, microSD, etc.)
- **Cloud Storage**

- Google Drive
- Dropbox
- OneDrive
- Amazon S3
- **Hybrid Storage**
 - Hybrid Drive (HDD + SSD Combination)
 - SSHD (Solid-State Hybrid Drive)

Cache Memory

Cache memory is smaller and faster than RAM. It is placed closer to the CPU than the RAM.



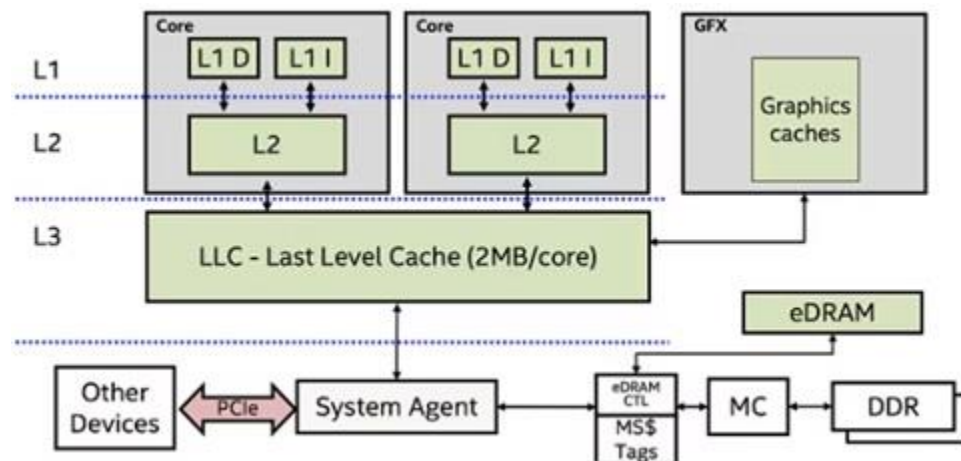
Different types of cache memory, like L1, L2, and L3 cache, have different speeds and amounts of space.

The Levels of Cache Memory: L1, L2, and L3

CPU Cache memory is divided into three "levels": L1, L2, and L3. The memory hierarchy is again according to the speed and, thus, the cache size.

L1 Cache

Level 1 cache is a computer's fastest memory. The CPU's most frequently accessed data resides in the L1 cache. CPU determines L1 cache size. Some high-end consumer CPUs, such as the Intel i9-9980XE, have a 1MB L1 cache, but they are expensive and rare. Server chipsets like Intel's Xeon have 1-2MB L1 memory cache. Before buying, examine the CPU specs to ascertain the L1 cache size. There is no "standard" amount.



Source: [1]

The L1 cache normally has two sections: the instruction cache, which stores CPU operation information, and the data cache, which stores operation data.

L2 Cache

Level 2 cache is larger but slower than L1. Modern L2 memory caches are gigabytes, not kilobytes. AMD's top-rated Ryzen 5 5600X has 384KB L1 and 3MB L2 caches and 32MB L3 cache. The L2 cache size depends on the CPU but is usually 256KB to 32MB. Nowadays, most CPUs have more than 256KB L2 cache, which is small. Some of the most powerful current CPUs have L2 memory caches exceeding 8MB. In terms of speed, the L2 cache is slower than the L1 cache but still faster than the system RAM. L2 caches are 25 times faster than RAM, while L1 caches are 100 times faster.

L3 Cache

Level 3 cache. The L3 memory cache was originally on the motherboard. This was long ago when most CPUs were single-core. The L3 cache on top-end consumer CPUs can reach 32MB, while AMD's groundbreaking Ryzen 7 5800X3D CPUs have 96MB. CPU L3 caches in some servers can reach 128MB.

The largest and slowest cache memory unit is L3. Modern CPUs have an on-chip L3 cache. The chip's L1 and L2 caches serve each core, while the L3 cache is more like a memory pool for the whole chip. The following images illustrate the CPU memory cache levels for a 2012 Intel Core i5-3570K CPU and a 2020 AMD Ryzen 5800X CPU. The second image's bottom right corner contains CPU cache data.



Source: [1]

Note how both CPUs have a split L1 cache and larger L2 and L3 caches. On the AMD Ryzen 5800X, the L3 cache is over five times greater than the Intel i5-3570K.

How cache memory works:

- **Hierarchy** – Computers normally have L1, L2, and L3 caches are the several layers of cache memory. The L1 cache is the smallest and fastest cache, located closest to the CPU; L2 and L3 caches are larger and slower.
- **Cache Organization** – Each block or line of cache memory contains a small bit of data copied from the main memory. The CPU accesses cache memory in fixed-size blocks, not bytes.
- **Cache Coherency** – Cache coherency ensures cached data matches the main memory data. Cache coherence techniques update other cores' caches when one core writes to a memory location in a multi-core processor.

- **Cache Replacement Policies** – A cache replacement policy decides which block to evict when the cache is full and a new block is needed. LRU, FIFO, and Random Replacement are common policies.
- **Cache Access** – The CPU checks the cache before reading or writing data. When data is cached, the CPU can quickly retrieve it. If data is not in the cache (cache miss), the CPU must fetch it from the main memory, which may delay it.
- **Cache Hierarchy** – Modern processors contain L1, L2, and L3 caches that grow in capacity and latency farther from the CPU cores. Parallel access is achieved by splitting the L1 cache into instruction and data caches.
- **Cache Management** – Optimization of cache utilization maximizes hit rates and minimizes miss penalties. Prefetching, where the processor predicts memory accesses and loads data into the cache, improves cache performance.

Cache memory buffers frequently access data between the CPU and main memory to speed up processing and increase system performance. Modern computer systems require effective management and structure for optimal performance.

Register Memory

Register memory, which is also called processor registers or "registers," is the smallest and fastest type of computer memory that is directly integrated into the CPU. Registers are small, fast storage units inside the CPU that are used to quickly store data that is being processed or instructions that are being run.

Registers serve several important functions in a computer system:

- **Instruction Execution** – Registers hold the instructions that the CPU is currently running. This includes the operation code (opcode) and associated operands with it.
- **Data Storage** – Registers store CPU-processed data. This can provide memory addresses, intermediate values during arithmetic or logical operations, and other data needed by the instructions being executed.
- **Addressing** – Memory addresses are used to store or retrieve data from memory locations in RAM or other parts of the computer's memory hierarchy.

Types of Registers

- **Program Counter (PC)** – Stores the memory address of the next instruction to be fetched and executed.
- **Instruction Register (IR)** – Holds the current instruction being executed by the CPU.
- **Memory Address Register (MAR)** – Stores the memory address of data being read from or written to memory.
- **Memory Data Register (MDR)** – Contains the actual data being read from or written to memory.
- **General-Purpose Registers (GPRs)** – Used for general data storage and manipulation during program execution.

Unit 2: Computer - Primary Memory

Computer - Primary Memory

Memory is required in computers to store data and instructions. Memory is physically organized as a large number of cells that are capable of storing one bit each. Logically they are organized as groups of bits called **words** that are assigned an address. Data and instructions are accessed

through these **memory address**. The speed with which these memory addresses can be accessed determines the cost of the memory. Faster the memory speed, higher the price.

Computer memory can be said to be organized in a hierarchical way where memory with the fastest access speeds and highest costs lies at the top whereas those with lowest speeds and hence lowest costs lie at the bottom. Based on this criteria memory is of two types **primary** and **secondary**. Here we will look at primary memory in detail.

What is Primary Memory?

The **primary memory** (or **main memory**) is a part of computer memory that can be directly accessed by the computer CPU. The **primary memories** are faster and store data temporarily. The data stored in the **primary memory** is erased when the computer is OFF. Thus, these memories are also known as **volatile memory**.

Features of Primary Memory

The main features of primary memory, which distinguish it from secondary memory are –

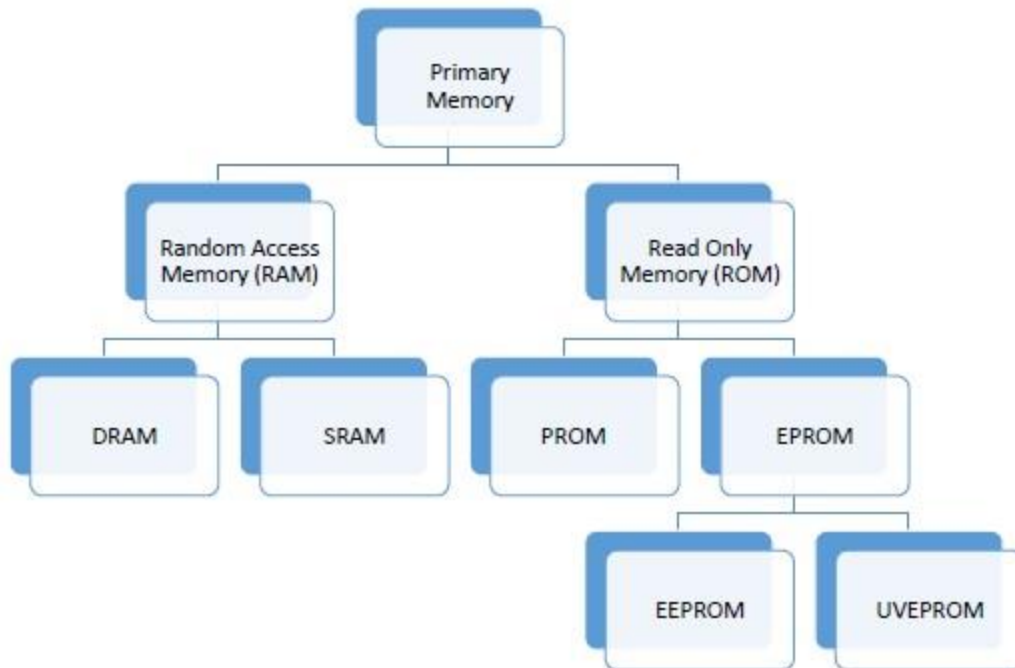
- It is accessed directly by the processor
- It is the fastest memory available
- Each word is stored as well as
- It is volatile, i.e. its contents are lost once power is switched off

As primary memory is expensive, technologies are developed to optimize its use. These are broad types of primary memory available.

Types of Primary Memory

The following are the main types of primary memory:

- RAM
- ROM
- Cache Memory



1. RAM (Random Access Memory)

RAM stands for **Random Access Memory**. The processor accesses all memory addresses directly, irrespective of word length, making storage and retrieval fast. RAM is the fastest memory available and hence most expensive. These two factors imply that RAM is available in very small quantities of up to 1GB. RAM is volatile but may be of any of these two types

i. DRAM (Dynamic RAM)

Each memory cell in a DRAM is made of one transistor and one capacitor, which store one bit of data. However, this cell starts losing its charge and hence data stored in less than thousandth of a second. So it needs to be refreshed thousand times a second, which takes up processor time. However, due to small size of each cell, one DRAM can have large number of cells. Primary memory of most of the personal computers is made of DRAM.

ii. SRAM (SRAM)

Each cell in SRAM is made of a flip flop that stores one bit. It retains its bit till the power supply is on and doesn't need to be refreshed like DRAM. It also has shorter read-write cycles as compared to DRAM. SRAM is used in specialized applications.

iii. VRAM (Video Random-Access Memory)

Video Random-Access Memory (VRAM) is a type of memory that is intended to work with video cards and graphics processing units (GPUs). It's a special place in memory where graphics data like images, frame buffers, and other graphics-related data can be stored.

VRAM is designed to handle the fast, parallel processing demands of rendering graphics and images on computer displays. It enables GPUs to quickly access large amounts of graphic data, which lets them render complex scenes, textures, and animations.

2. ROM (Read Only Memory)

ROM stands for **Read Only Memory**. As the name suggests, ROM can only be read by the processor. New data cannot be written into ROM. Data to be stored into ROM is written during the manufacturing phase itself. They contain data that does not need to be altered, like booting

sequence of a computer or algorithmic tables for mathematical applications. ROM is slower and hence cheaper than RAM. It retains its data even when power is switched off, i.e. it is non-volatile. ROM cannot be altered the way RAM can be but technologies are available to program these types of ROMs –

i. PROM (Programmable ROM)

PROM can be programmed using a special hardware device called PROM programmer or PROM burner.

ii. EPROM (Erasable Programmable ROM)

EPROM can be erased and then programmed using special electrical signals or UV rays. EPROMs that can be erased using UV rays are called UVEEPROM and those that can be erased using electrical signals are called EEPROM. However, handling electric signals is easier and safer than UV rays.

iii. EEPROM (Electrically Erasable Programmable Read Only Memory)

Information is written by the user using an EPROM programmer. Unlike EPROM, EEPROM data is erased using electrical signals rather than UV light. EEPROM allows selective data erasure and is more expensive than other ROM types. It is gaining popularity due to its flexibility.

3. Cache Memory

The small piece of high speed volatile memory available to the processor for fast processing is called cache memory. Cache may be a reserved portion of main memory, another chip on CPU or an independent high speed storage device. Cache memory is made of fast speed SRAMs. The process of keeping some data and instructions in cache memory for faster access is called **caching**. Caching is done when a set of data or instructions is accessed again and again.

Whenever the processor needs any piece of data or instructions, it checks the cache first. If it is unavailable there, then the main memory and finally secondary memory is accessed. As cache has very high speed, time spent in accessing it every time is negligible as compared to time saved if data indeed is in the cache. Finding data or instruction in cache is called **cache hit**.

Unit 3: Computer - Secondary Memory

You know that processor memory, also known as primary memory, is expensive as well as limited. The faster primary memory are also volatile. If we need to store large amount of data or programs permanently, we need a cheaper and permanent memory. Such memory is called **secondary memory**. Here we will discuss secondary memory devices that can be used to store large amount of data, audio, video and multimedia files.

What is Secondary Memory?

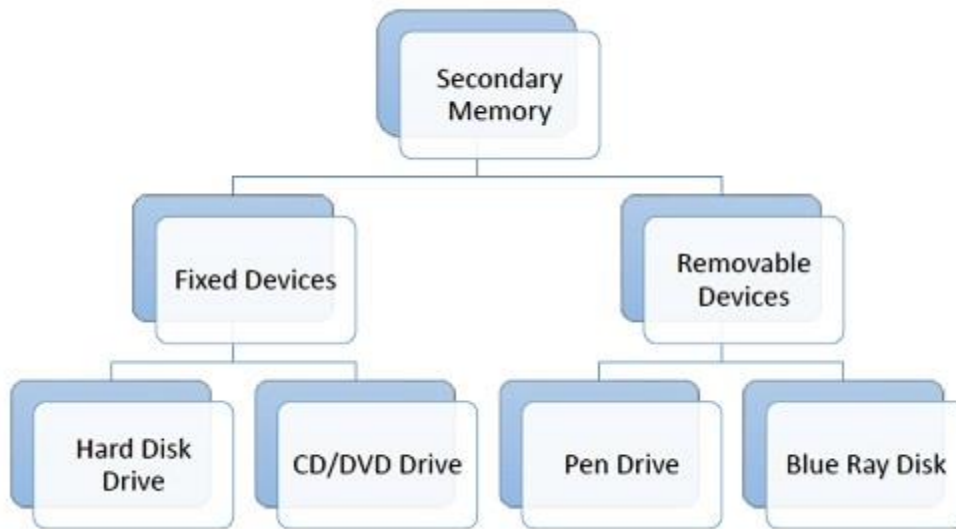
The **secondary memory** (or **external memory**) is a part of computer memory that cannot be directly accessed by the computer CPU. The **secondary memories** are slower than primary memories and store data permanently. The secondary memory is also known as **non-volatile** or **auxiliary memory**.

Characteristics of Secondary Memory

These are some characteristics of secondary memory, which distinguish it from primary memory

–

- It is non-volatile, i.e. it retains data when power is switched off
- It is large capacities to the tune of terabytes
- It is cheaper as compared to primary memory



Types of Secondary Memory

- **Magnetic Storage**
 - Hard Disk Drive (HDD)
 - Floppy Disk (Obsolete)
 - Magnetic Tape
- **Optical Storage**
 - CD (Compact Disc)
 - CD-ROM (Read-Only Memory)
 - CD-R (Recordable)
 - CD-RW (Rewritable)
 - DVD (Digital Versatile Disc)
 - DVD-ROM
 - DVD-R/DVD+R
 - DVD-RW/DVD+RW
 - Blu-ray Disc (BD)
- **Flash Storage (Solid-State Storage)**
 - Solid-State Drive (SSD)
 - USB Flash Drive (Pen Drive)
 - Memory card (SD Card, microSD, etc.)
- **Cloud Storage**
 - Google Drive
 - Dropbox
 - OneDrive
 - Amazon S3
- **Hybrid Storage**
 - Hybrid Drive (HDD + SSD Combination)
 - SSHD (Solid-State Hybrid Drive)

Let us look at some of the secondary memory devices available.

Hard Disk Drive

Hard disk drive is made up of a series of circular disks called **platters** arranged one over the other almost inches apart around a **spindle**. Disks are made of non-magnetic material like aluminum alloy and coated with 10-20 nm of magnetic material.



Standard diameter of these disks is 14 inches and they rotate with speeds varying from 4200 rpm (rotations per minute) for personal computers to 15000 rpm for servers. Data is stored by magnetizing or demagnetizing the magnetic coating. A magnetic reader arm is used to read data from and write data to the disks. A typical modern HDD has capacity in terabytes (TB).

CD Drive

CD stands for **Compact Disk**. CDs are circular disks that use optical rays, usually lasers, to read and write data. They are very cheap as you can get 700 MB of storage space for less than a dollar. CDs are inserted in CD drives built into CPU cabinet. They are portable as you can eject the drive, remove the CD and carry it with you. There are three types of CDs –

- **CD-ROM (Compact Disk Read Only Memory)** – The data on these CDs are recorded by the manufacturer. Proprietary Software, audio or video are released on CD-ROMs.
- **CD-R (Compact Disk Recordable)** – Data can be written by the user once on the CD-R. It cannot be deleted or modified later.
- **CD-RW (Compact Disk Rewritable)** – Data can be written and deleted on these optical disks again and again.

DVD Drive

DVD stands for **Digital Video Display**. DVD are optical devices that can store 15 times the data held by CDs. They are usually used to store rich multimedia files that need high storage capacity. DVDs also come in three varieties read only, recordable and rewritable.



Pen Drive

Pen drive is a portable memory device that uses solid state memory rather than magnetic fields or lasers to record data. It uses a technology similar to RAM, except that it is nonvolatile. It is also called USB drive, key drive or flash memory.



Blu Ray Disk

Blu Ray Disk (BD) is an optical storage media used to store high definition (HD) video and other multimedia files. BD uses shorter wavelength laser as compared to CD/DVD. This enables writing arm to focus more tightly on the disk and hence pack in more data. BDs can store up to 128 GB data.

Computer - Random Access Memory (RAM)

What is RAM (Random Access Memory)?

RAM is a computer's volatile memory which stores data temporarily; once power turned off, the data which is inside the RAM gets erased. It is also known as primary or temporary memory. RAM stores the data that is actively being used during the execution of a program or instructions. During the booting process of a system, the operating system actively uses RAM and applications that are

necessary to execute a file or a program. It speeds up CPU processing by providing fast data and instruction access.



RAM is actively used for program or instruction execution. Once we start the computer; system necessary files, programs and operating system files are loaded into the RAM for the smooth running of the computer. The more RAM a computer has, the better it can handle multitasking and the faster it can run applications since data can be accessed more quickly.

How does RAM Work?

RAM consists of a grid of cells, each with a unique address, allowing data to be read or written in any order, the CPU requests data from RAM, which is delivered at high speed, making it essential for multitasking and running complex applications.

RAM is the computer's internal memory which stores different types of data and information as per the requirement. It is also considered as read/write memory. The data access in RAM is random. This means that we can access any random location in the memory without even knowing the address of the previous location. Since RAM is a volatile memory hence, a backup is present in most of the systems in the form of an uninterrupted power supply (UPS). Moreover, the speed and performance of a system are directly proportional to the size of the RAM.

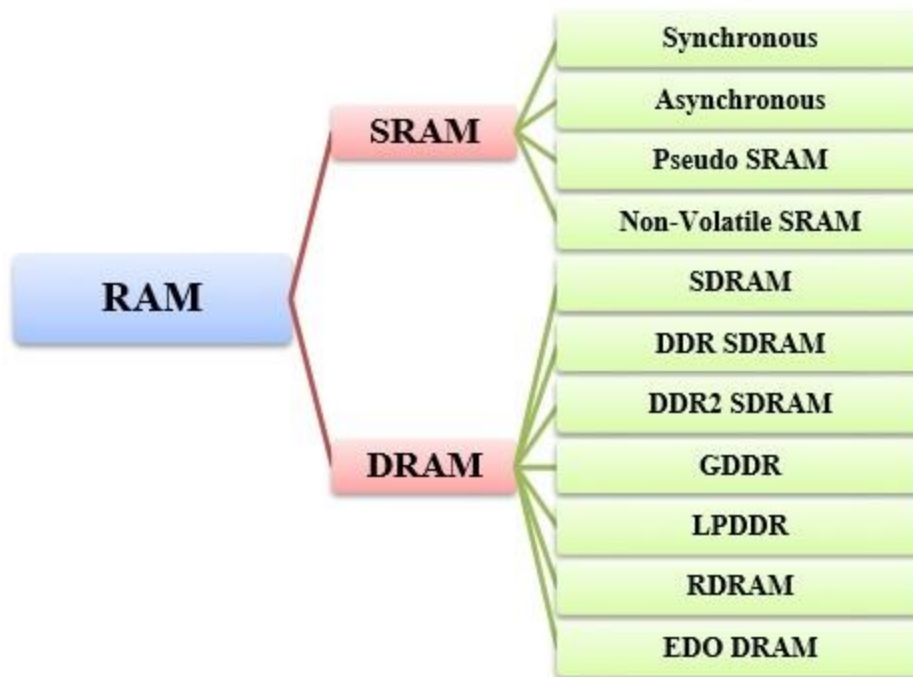
Characteristics of RAM

Key Characteristics of RAM are described below –

- **Multitasking** – RAM does multitasking and fast execution of instructions or programs.
- **Prevents system crashes** – RAM prevents system crashes and slowdowns during intensive tasks.
- **High-performance** – It provides high performance for the execution of gaming and visual applications, where large amounts of data must be processed swiftly.
- **System Support** – RAM can perform write and erase operations.
- **Additional RAM** – To maintain the systems speed and performance; additional RAM can be added on.
- **Faster data accessing** – It facilitates CPU by accessing data quickly to support faster execution. RAM is faster than SSD.
- **Cost** – RAM is not costly as compared to SSD.

Types of Random Access Memory

Different types of RAM are categorised in the below image



SRAM (Static Random Access Memory)

Static RAM is a type of RAM that uses flip-flops for primary storage and is volatile. It is a type of semiconductor memory. Each bit is stored using bistable latching circuitry, making it extremely fast so known as cache memory. It is commonly used as a CPU cache and is typically located in L2 or L3. It stores data in the latch as long as the computer is on. SRAM is more expensive and requires more power than DRAM. It serves as cache memory in a computer system. SRAM requires more transistors than DRAM. It is faster than DRAM because of the latching mechanism. SRAM employs 6 transistors per data bit while DRAM uses one transistor per bit.

Characteristics of SRAM

Some of the key characteristics of SRAM are as follows –

- SRAM has a memory cell which consists of 6 transistors.
- SRAM consist flip-flop to store data units.
- SRAM is dedicatedly designed for the CPU and works as a cache memory.
- It accesses data in less time; faster than DRAM, and takes very less time to access data.
- Storing a large amount of data on a single chip is not possible, so more chips are needed.
- SRAM is expensive.
- It requires a regular power supply to store data.
- SRAM generates more heat.
- Contents remain as it is until power is ON.

Types of SRAM

SRAM can be further classified into the following types –

- **Synchronous** – Synchronous SRAM works jointly with a clock signal. It reads and writes in sync with the clock, making it faster and suitable for high-speed applications.
- **Asynchronous** – Its working is not dependent on the clock signal; it carries out read-and-write operations once the addresses and control signals are stable.

- **Pseudo SRAM** – PSRAM is a type of memory that has features of both DRAM and SRAM. The device uses a DRAM cell but includes an internal refresh circuit, so presenting itself as SRAM to the external system.
- **Non-Volatile SRAM** – nvSRAM integrates a small non-volatile memory with SRAM cells to keep data even when power is off.

Advantages of SRAM

The advantages of SRAM are as follows –

- SRAM is preferred because it is much faster.
- It does not require being refreshed.
- It delivers faster read and write operations.
- It works as a cache memory in CPUs.
- It is less prone to errors and data corruption caused by the refresh process.
- It is very reliable and therefore is used for cache memory.

Disadvantages of SRAM

Disadvantages of DRAM are as follows –

- It is expensive.
- It has a complex cell structure; so the design and layout of SRAM are more complex.
- Due to its volatile nature, overall data is lost when power goes off.
- It has a low storage capacity as compared to DRAM.
- The design is complex and not easy to build or understand.
- It is not Suitable for Large Memory Applications

Overall, speed and reliability make SRAM highly-priced, particularly for applications such as CPU caches. However, its higher cost and lower density compared to DRAM restrict its use to smaller, high-performance memory applications.

DRAM (Dynamic Random Access Memory)

It is another type of RAM used as main memory; it stores data in capacitors for a short period (a few milliseconds) even when the computer is powered on. Each memory cell is a DRAM chip that contains one bit of data and is composed of transistors and capacitors. In this memory controller, the data can be read and then rewritten, constantly updating it. The data is refreshed regularly to ensure its integrity. The DRAM is less expensive, but it can store significantly more data. In addition, it is slower and requires less power than SRAM.

However, DRAM is less expensive than SRAM, making it suitable for use as the CPU's main memory. It is slower than SRAM and can link directly to the CPU bus. A DRAM typically ranges from 4GB to 16GB for laptops and 1GB to 2GB for smaller devices.

Characteristics of DRAM

Some of the key characteristics of DRAM are as follows –

- It works as a main memory for the system.
- It is made up of capacitors that store the data, thus we need to refresh it multiple times.
- It is inexpensive so DRAM is mostly available in Laptops or PCs today.
- It requires less space and less power.
- It refreshes every millisecond to maintain data consistency.
- It is more power-efficient than SRAM.
- DRAM can store multiple bits on a single chip.

Types of DRAM

DRAM can be further classified into the following types –

- **SDRAM** – Synchronous with bus clock; general computing.

- **DDR SDRAM** – it has higher data transfer rates compared to SDRAM.
- **DDR2 SDRAM** – It has higher speeds, lower power consumption, and enhanced latency.
- **GDDR** – It is optimized for graphics processing units (GPUs).
- **LPDDR** – It is dedicatedly designed for mobile and embedded applications.
- **RDRAM** – Specifically designed for innovative design for specific high-performance applications.
- **EDO DRAM** – It allows for faster read cycles by overlapping operations.

Advantages of DRAM

The advantages of DRAM are as follows –

- DRAM has a slower accessing time than SRAM
- Its design is simple, not complex because it consists of only a single transistor.
- It is smaller in size.
- Its cost is low compared to SRAM
- It has high congestion rates.
- It has a large memory space.
- It has a high density that makes it suitable for large memory applications.

Disadvantages of DRAM

Disadvantages of DRAM are as follows –

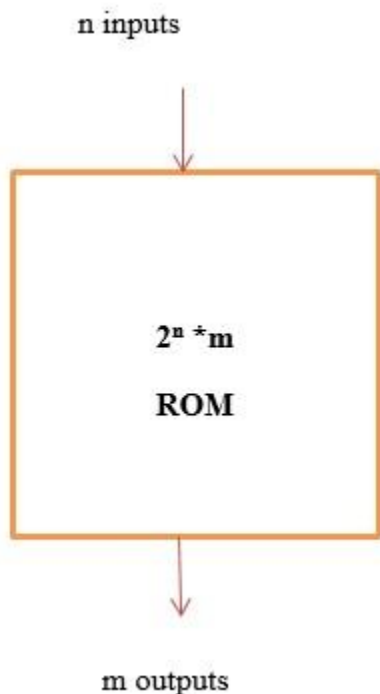
- It is volatile so continuous refreshing requires keeping data consistent.
- Its manufacturing process is complex.
- It is slower than SRAM.
- It has higher latency compared to SRAM.

MODULE 8: COMPUTER - READ ONLY MEMORY (ROM)

Unit 1: Read Only Memory (ROM)

What is ROM?

ROM is a non-volatile memory that stores data permanently. Advanced ROM data can be erased and rewritten. The information in ROM can only be accessed, making it perfect for use in computers as it enables quick access to data without the need for constant data rewriting. Important data, like BIOS settings and operating system instructions, are stored in these memories, which are crucial for the proper functioning of your computer. Without them, your computer would essentially be "brain dead," so they must continue to work properly at all times!



ROM is an efficient storage method requiring minimal effort and maintenance, capable of holding instructions and various types of data.

Advertisement

Characteristics of Read Only Memory (ROM)

The primary characteristics of ROM are as follows –

- ROM is Non-volatile in nature.
- Advanced ROM can be rewritten.
- Data or information stored on ROM can be read-only.
- ROM holds its data even when power is turned off.
- The read-only characteristic ensures data stability and integrity.
- ROM utilization occurs during the computer's boot-up procedure.

How does Read Only Memory (ROM) Work?

ROM functions by using memory cells, which are composed of transistors arranged in series to enable the flow of electrical current when powered. This current passing through the transistors generates a magnetic field that retains the data in the memory cell until it is erased by another

signal passing through the transistor. The information stored in these cells cannot be altered or deleted unless directed by an external entity such as a computer processor or hard drive controller. Overall, inside the computer, a small but durable battery powers the ROM. Its two essential components are the decoder and OR logic gates. Binary input from the ROM is processed by the decoder, which then produces the decimal equivalent as output. The decimal output of the decoder is used as the input for the OR gates in the ROM.

ROM is akin to an array of disks. The system can be activated and deactivated using a grid of rows and columns. Each element of the array is assigned to a specific memory element on the ROM. A diode is used to connect the relevant components. When a request is sent, the address provided is used to locate the specific memory location. The value extracted from the ROM chip must match the value in the array's selected element.

Types of Read-Only Memory (ROM)

Based on its characteristics and uses, ROM is broadly classified into four different types –

1. MROM (Masked Read-Only Memory)

MROM stands for Masked Read-Only Memory. This memory device is no longer used nowadays due to obsolescence. Manufacturers are using this memory to store software and information during the manufacturing process. Therefore, it is impossible to alter, reprogram, or erase this type of pre-programmed chip later.

Applications of MROM

It is most widely used in Network Operating Systems and Server Operating Systems.

2. PROM (Programmable Read-Only Memory)

PROM stands for Programmable Read-Only Memory. Generally, it's a blank memory chip. It can be programmed during manufacturing and cannot be changed or erased. A PROM programmer or PROM burner is used to write information to a PROM. The process of programming a PROM chip is commonly known as "PROM Burning." Once programmed, the chip's content cannot be changed. As a result, it is also known as a one-time programmable device.

Applications of PROM

It is most widely used in Computer Bios, TV Remote, Video game consoles and nowadays a day used in Mobile Phones.

3. EPROM (Erasable Programmable Read-Only Memory)

EPROM stands for Erasable Programmable Read-Only Memory. It can be programmed and later erased by exposing it to strong ultraviolet light. After being erased, the EPROM can be reprogrammed. EPROM is a type of read-only memory chip that may erase previously programmed data. High voltage can be used to program the programmable read-only memory to write data, which is then retained until exposed to ultraviolet light for 10 minutes or more.

Applications of EPROM

It is most widely used in Video cards, Memory sticks, BIOS chips, and Modem.

4. EEPROM (Electrically Erasable Programmable Read-Only Memory)

EEPROM stands for Electrically Erasable Programmable Read-Only Memory. It is an advanced type of ROM that can be erased and reprogrammed electrically. This operation can be repeated indefinitely, and the computer system can do it while running. The EEPROM allows users to program and erase any desired spot. It is erased byte by byte rather than the full chip. As a result, reprogramming is flexible and takes less time to complete the process.

Applications of EEPROM

It is most widely used in Bios Chip and Microcontrollers for Smart Cards.

5. Flash Memory

It is a type of EEPROM that can be erased and reprogrammed in blocks rather than one byte at a time which allows for faster and more efficient operations. It is widely employed as storage in modern devices such as USB drives, memory cards, and SSDs. A modern variant of EEPROM is flash memory. With its newer designs and high endurance, it may be erased and rewritten faster than typical EEPROMs.

Applications of Flash Memory

It is most widely used in storage devices like USB drives, memory cards, and SSDs.

Advantages of Using (Read Only Memory) ROM

The primary advantages of ROM are as follows –

- It is non-volatile and less expensive than RAM.
- Its circuit design is simple as compared to RAM.
- No need to refresh it.
- ROM provides fast data access.
- It is reliable.
- Data remains intact and uncorrupted.
- It maintains data integrity and security.

Applications of Read Only Memory (ROM)

There are various applications of ROM. Some of those applications are as follows –

- **Supports the Booting Process** – It supports the booting process of the computer and loads the operating system.
- **Input/output (I/O) Tasks** – ROM performs large I/O tasks.
- **Stores Data** – ROM stores data and updates firmware software.
- **Embedded Systems** – ROMs are most widely used in microwaves, washing machines, and related appliances where firmware controls operations.
- **Consumer Electronics** – It is used in modern smartphones for storing bootloaders and firmware.
- **Computer BIOS** – It provides support for basic Input / Output System (BIOS) in PCs.

ROM is an essential memory device in different electronic systems due to its impact use like reliability, security, and cost-effectiveness.

RAM Vs. ROM

The primary difference between RAM and ROM are as –

RAM	ROM
It stands for Random Access Memory	It stands for Read-only Memory
It is more expensive than ROM	It is less expensive than RAM
Its speed is high	Its speed is low

Temporary storage for data and programs	Permanent storage for data and programs
Its storage capacity is high compared to ROM	Its storage capacity is low compared to the RAM
It can perform Read, Modify and Delete operations	It performs only Read operations
It performs processing on current instructions	It performs Bootstrap processing
RAM is a fast	ROM is slow
Probably used in the CPU cache and primary memory	Probably used in firmware and micro-controllers
It is volatile	It is not volatile

Dynamic Random Access Memory (DRAM)

Dynamic RAM (DRAM)

Dynamic RAM (DRAM) is a type of semiconductor memory that uses capacitors to store the bits. The charging and discharging of the capacitor represents 0 and 1 i.e. the two possible values that can be stored in a bit.

The DRAM is a volatile memory i.e. the data in memory is lost when power is switched off. However, it still displays some data remanence. DRAM is low cost compared to SRAM so it is primarily used in main memory.

The following displays a DRAM:



Versions of DRAM

Asynchronous DRAM

This was the first type of DRAM in use but was gradually replaced by synchronous DRAM. This was called asynchronous because the memory access was not synchronized with the system clock.

Synchronous DRAM

This DRAM replaced the asynchronous RAM and is used in most computer systems today. In synchronous DRAM, the clock is synchronised with the memory interface. All the signals are processed on the rising edge of the clock.

Graphics DRAM

There are many graphics related tasks that can be accomplished with both synchronous and asynchronous DRAM. Some of the DRAM used for these tasks are Video DRAM, Window DRAM, Multibank DRAM etc.

Difference between DRAM and SRAM

There are many differences between DRAM and SRAM. Some of these are given as follows:

- Both static and dynamic RAM are types of RAM but SRAM is formed using flip flops and DRAM using capacitors.
- It is necessary that the data in DRAM is refreshed periodically to store it correctly. This is not necessary for SRAM.
- SRAM is normally only used in Cache memory while DRAM is used in main memory.
- Static RAM is much more faster and expensive as compared to Dynamic RAM.
- Since SRAM is used as Cache memory, its size is 1MB to 16MB. On the other hand, dynamic memory is larger as it is used as main memory. Its size is 4GB to 16GB in computers and laptops.
- SRAM is usually present on processors or between processors and main memory. DRAM is present on the motherboard.

Unit 2: Computer - Video Random-Access Memory (VRAM)

Computer - Video Random-Access Memory (VRAM)

What is Video Random-Access Memory (VRAM)?

Video Random-Access Memory (VRAM) is a type of memory that is dedicated to computers video cards and graphics processing units (GPUs). It's specifically designed to store and process graphics and video data like images, and videos.

VRAM is designed to handle the fast, parallel processing demands of rendering graphics and images on computer displays. It enables GPUs to quickly access large amounts of graphic data, which lets them render complex scenes, textures, and animations.

The video memory is located on the graphics card or GPU and is distinct from the computer's RAM. It is specially made to meet the high bandwidth requirements of graphics-heavy applications.

Components of Video Random-Access Memory (VRAM)

VRAM is comprised of several memory banks or modules, enabling it to access data simultaneously, resulting in enhanced performance. The primary variants of VRAM are Graphics Double Data Rate (GDDR) and High-Bandwidth Memory (HBM), each providing distinct benefits in terms of speed and power efficiency.

Why Video Random-Access Memory (VRAM) is important?

Video memory is essential in gaming because it stores and handles the textures, shaders, and other graphical data required to generate realistic and immersive gaming worlds. Sufficient video memory makes sure smooth gameplay without graphical distortions or lag. Video memory has a direct impact on graphics performance since it determines how much data the graphics processing

unit can store and access. Insufficient video memory can result in slower frame rates, lower resolution textures, and a general decrease in graphics quality.

How does Video Random-Access Memory (VRAM) Work?

VRAM operates on a parallel architecture, allowing it to read and write data simultaneously. This enables faster data access and manipulation, which is crucial for real-time rendering. The VRAM's high bandwidth and low latency contribute to delivering smooth and lag-free graphics performance.

Characteristics of Video Random-Access Memory (VRAM)

The primary characteristics or features of VRAM include the followings –

- **High Bandwidth** – VRAM typically offers high-speed data transfer rates, enabling fast access to graphical data by the GPU.
- **Fast** – VRAM is fast and has a high bandwidth, making it perfect for graphics devices.
- **Parallel Access** – VRAM is designed to support parallel access, allowing multiple rendering tasks to access different portions of the memory simultaneously.
- **Specialized Architecture** – VRAM often has a specialized architecture optimized for graphics processing tasks, including features such as multi-port access and wide memory buses.
- **Dedicated Graphics Memory** – Unlike system RAM, which is shared among various system components, VRAM is dedicated solely to graphics processing, ensuring that the GPU has sufficient memory bandwidth and capacity for rendering graphics-intensive applications.
- **Dual-Ported** – VRAM is Dual-Ported; refresh the current display content.

Types of Video Random-Access Memory (VRAM)

Types of VRAM are described below –

1. Window RAM (WRAM)

The RAM is extremely fast, dual-ported, and has approximately 25% more bandwidth than any regular VRAM. It is suitable for very high resolution (e.g., 1600X1200 px) with true colour.

2. Synchronous Graphics RAM (SGRAM)

The RAM is relatively inexpensive and synchronizes with the clock. It indicates that data can be edited with a single operation rather than a series of write, read, and update processes.

3. Rambus Dynamic RAM (RDRAM)

It incorporates a proprietary Rambus bus which speeds up data flow between the frame buffer and VRAM. RDRAM is an optimization for video streaming.

4. GDDR (Graphics Double Data Rate) VRAM

This is the most commonly used type of VRAM, it is majorly found in modern GPUs. GDDR5, GDDR5X, and GDDR6 are some of the variants that provide improvements in power efficiency and bandwidth over earlier generations.

5. Multibank Dynamic RAM (MDRAM)

MoSys developed this high-performance RAM. It separates the memory into 32KB segments that can be accessed independently. It improves the efficiency of memory transfers and the overall performance.

6. HBM (High Bandwidth Memory)

HBM VRAM is a more recent technology that provides even greater bandwidth at lower power consumption than conventional GDDR VRAM. It accomplishes this by minimising the distance data must travel between memory cells by stacking memory chips vertically on a silicon interposer.

Advantages of Video Random-Access Memory (VRAM)

Some of the key advantages of VRAM are as follows –

- VRAM is specifically designed to process graphics data at very high speed.
- It can handle complex image textures or render polygon-based 3D structures.
- It is most widely used in gaming performance.
- VRAM can produce more complex images and textures on graphics chips, which leads to better performance.
- VRAM ensures smooth and effective rendering.
- VRAM contributes significantly to faster loading times and higher image quality.

Unit 3: Computer - Cache Memory

What is Cache Memory?

Memory that is smaller and faster than RAM is called cache memory. It is a volatile memory fixed closer to the CPU to provide high-speed data access to the processor and stores frequently used computer programs, applications and data. Generally a cache memory fixes between processor and Random Access Memory.



What does Cache Memory Look Alike?

Cache memory's physical structure can be understood using the below images.



It saves data and instructions that are used a lot so that processing goes faster.

Why Cache Memory is Important?

Cache memory is much faster than main memory, thus the CPU can access data quickly used in execution; cache memory helps to get full utilisation of the CPU. It generally fixes closer to the CPU to communicate with CPU and RAM. Cache memory increases the system's overall performance and efficiency. It reduces the latency that the CPU experiences when retrieving data from main memory.

How does Cache Memory Work?

The data in a cache is typically stored in hardware with immediate access, like RAM. When a user retrieves data, it checks the caching first. If the data is located in the cache, it is called a cache hit. The proportion of attempts that provide a cache hit is known as the cache hit rate or ratio. Data that is not located in the cache is taken from the main memory and put into the cache. This is known as a cache miss. How this is accomplished and what information is expelled from the cache

to create space for new data is determined by the caching algorithms, cache mechanisms, and system regulations.

Different caching mechanisms govern the operation of the cache.

- **Cache Hierarchy** – Computers normally have L1, L2, and L3 caches are the several layers of cache memory. The L1 cache is the smallest and fastest cache, located closest to the CPU; L2 and L3 caches are larger and slower. Modern processors contain L1, L2, and L3 caches that grow in capacity and latency farther from the CPU cores. Parallel access is achieved by splitting the L1 cache into instruction and data caches.
- **Cache Organization** – Each block or line of cache memory contains a small bit of data copied from the main memory. The CPU accesses cache memory in fixed-size blocks, not bytes.
- **Cache Coherency** – Cache coherency ensures cached data matches the main memory data. Cache coherence techniques update other cores' caches when one core writes to a memory location in a multi-core processor.
- **Cache Replacement Policies** – A cache replacement policy decides which block to evict when the cache is full and a new block is needed. LRU, FIFO, and Random Replacement are common policies.
- **Cache Access** – The CPU checks the cache before reading or writing data. When data is cached, the CPU can quickly retrieve it. If data is not in the cache (cache miss), the CPU must fetch it from the main memory, which may delay it.
- **Cache Management** – Optimization of cache utilization maximizes hit rates and minimizes miss penalties. Prefetching, where the processor predicts memory accesses and loads data into the cache, improves cache performance.

Cache memory buffers frequently access data between the CPU and main memory to speed up processing and increase system performance. Modern computer systems require effective management and structure for optimal performance.

Types of Cache Memory

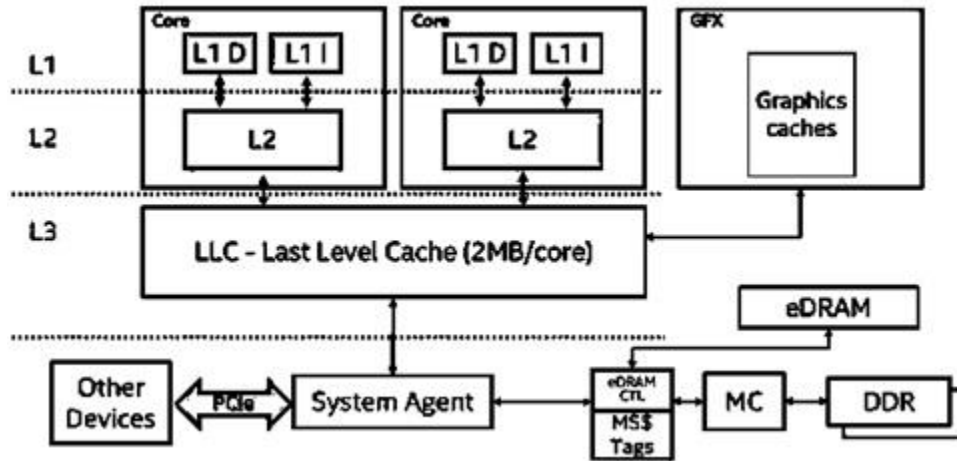
Different types of cache memory, like L1, L2, and L3 cache, have different speeds and amounts of space.

The Levels of Cache Memory: L1, L2, and L3

CPU Cache memory is categorised into three "levels": L1, L2, and L3.

L1 Cache

Level 1 cache is a computer's fastest memory. The CPU's most frequently accessed data resides in the L1 cache. CPU determines L1 cache size. Some high-end consumer CPUs, such as the Intel i9-9980XE, have a 1MB L1 cache, but they are expensive and rare. Server chipsets like Intel's Xeon have 1-2MB L1 memory cache. Before buying, examine the CPU specs to ascertain the L1 cache size. There is no "standard" amount.



The L1 cache normally has two sections: the instruction cache, which stores CPU operation information, and the data cache, which stores operation data.

L2 Cache

Level 2 cache is larger but slower than L1. Modern L2 memory caches are gigabytes, not kilobytes. AMD's top-rated Ryzen 5 5600X has 384KB L1 and 3MB L2 caches and 32MB L3 cache. The L2 cache size depends on the CPU but is usually 256KB to 32MB. Nowadays, most CPUs have more than 256KB L2 cache, which is small. Some of the most powerful current CPUs have L2 memory caches exceeding 8MB. In terms of speed, the L2 cache is slower than the L1 cache but still faster than the system RAM. L2 caches are 25 times faster than RAM, while L1 caches are 100 times faster.

L3 Cache

Level 3 cache. The L3 memory cache was originally on the motherboard. This was long ago when most CPUs were single-core. The L3 cache on top-end consumer CPUs can reach 32MB, while AMD's groundbreaking Ryzen 7 5800X3D CPUs have 96MB. CPU L3 caches in some servers can reach 128MB.

The largest and slowest cache memory unit is L3. Modern CPUs have an on-chip L3 cache. The chip's L1 and L2 caches serve each core, while the L3 cache is more like a memory pool for the whole chip. The following images illustrate the CPU memory cache levels for a 2012 Intel Core i5-3570K CPU and a 2020 AMD Ryzen 5800X CPU. The second image's bottom right corner contains CPU cache data.

Applications of Cache Memory

The primary applications of Cache Memory are as follows –

- Cache stores temporary files; so it is most widely used in the CPU.
- Different programs and software use their cache.
- A web browser has a cache that retains information.
- Caches are utilised by the operating system.
- It stores frequently accessed data which makes data access fast if a user re-accesses it.

Advantages of Cache Memory

Some key advantages of cache memory are as follows –

- **Faster Access** – It is closer to CPU so access data very fast.
- **Reducing Memory Latency** – It takes very little time to retrieve data from memory which shows it reduces memory latency.

- **Increasing effective CPU utilization** – It supports full CPU utilisation by fast data accessing and giving it to the CPU so that the execution of instructions or programs is done on time.
- **Reduced Power Consumption** – Cache memory consumes much less power than other types of storage.
- **Scalability** – Cache memory can grow with a system's needs.

Overall, a cache memory increases throughput, improves reliability and increases scalability.